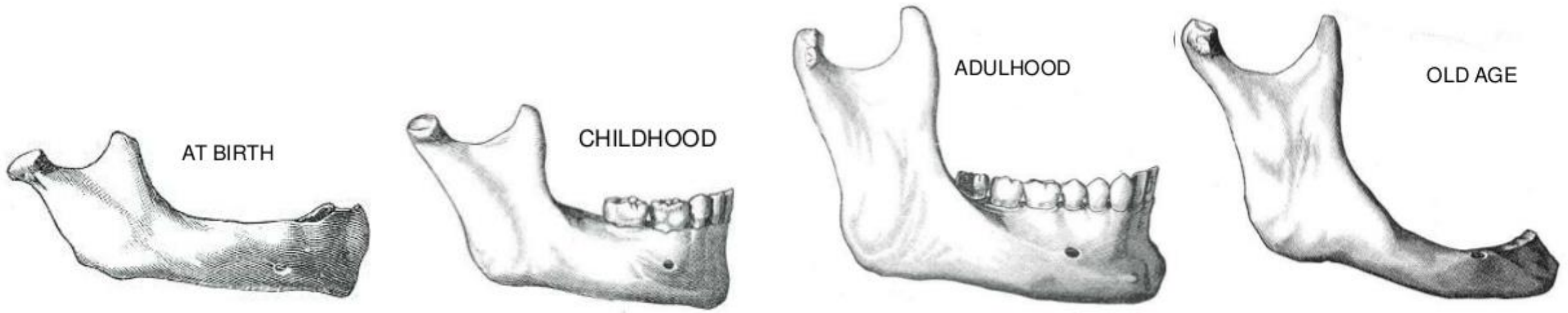


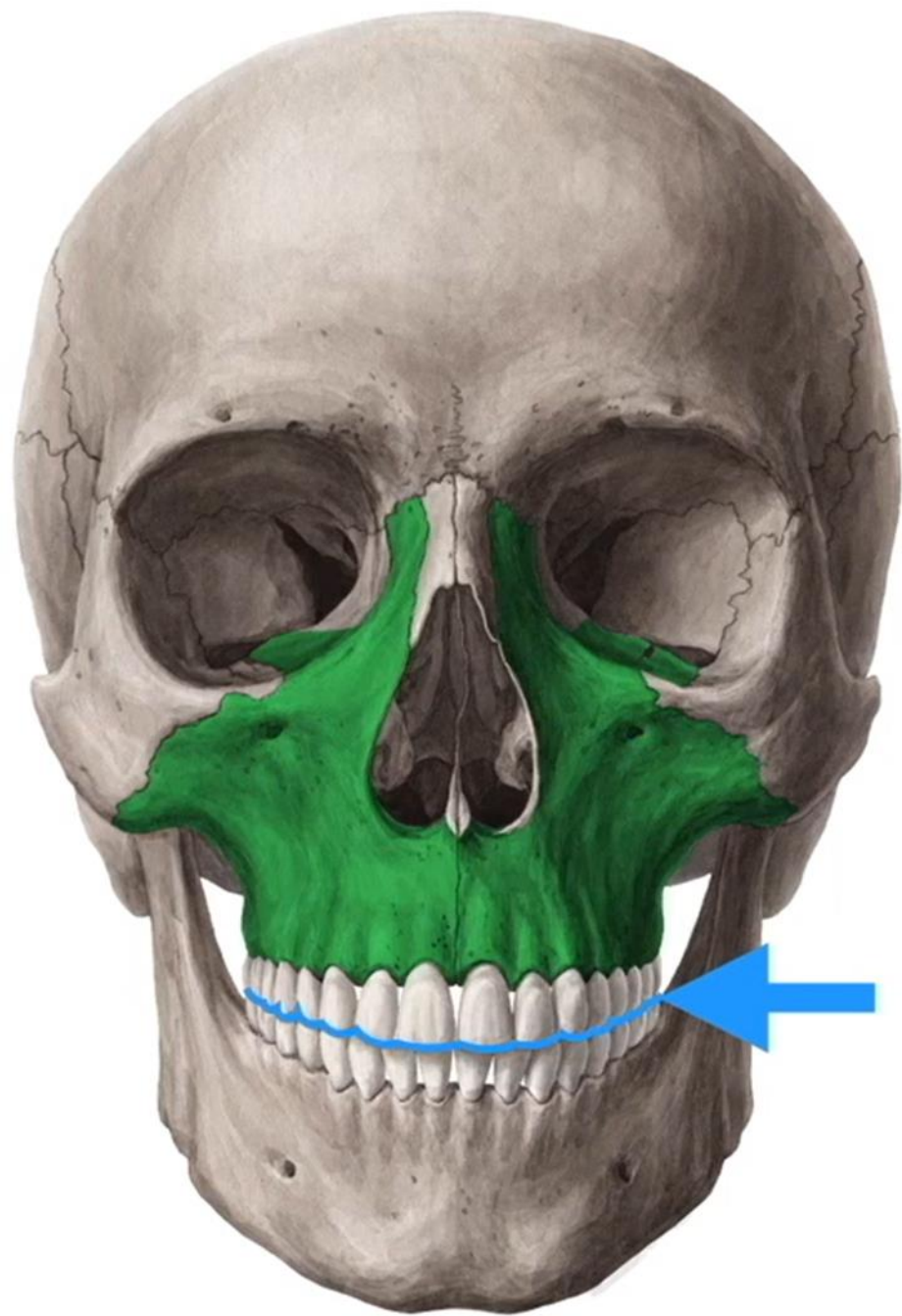
Mandible at different ages



By

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Lecturer of Oral biology
Cairo University



Mandible anatomy





Body



Ramus

Body of the mandible

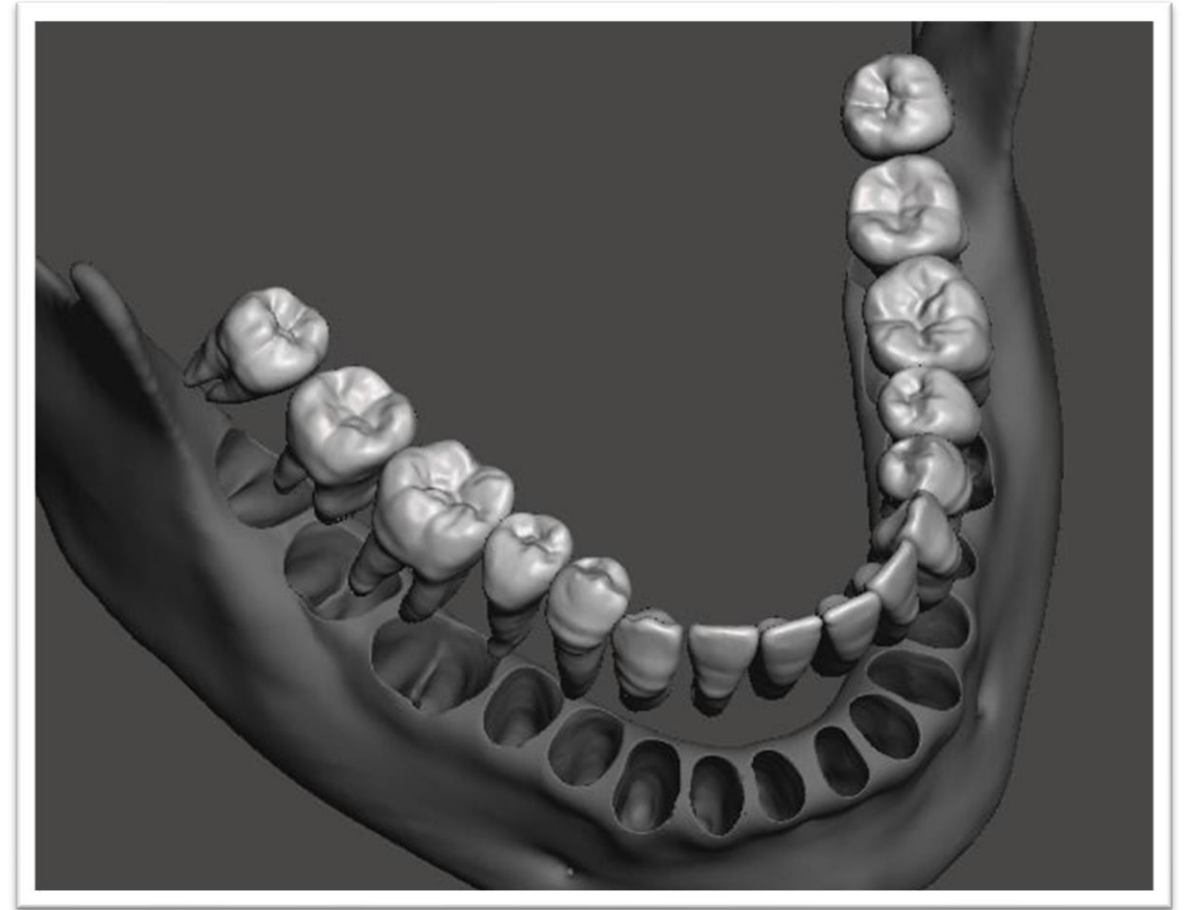


Alveolar bone



Lower border

Body of the mandible



Body of the mandible

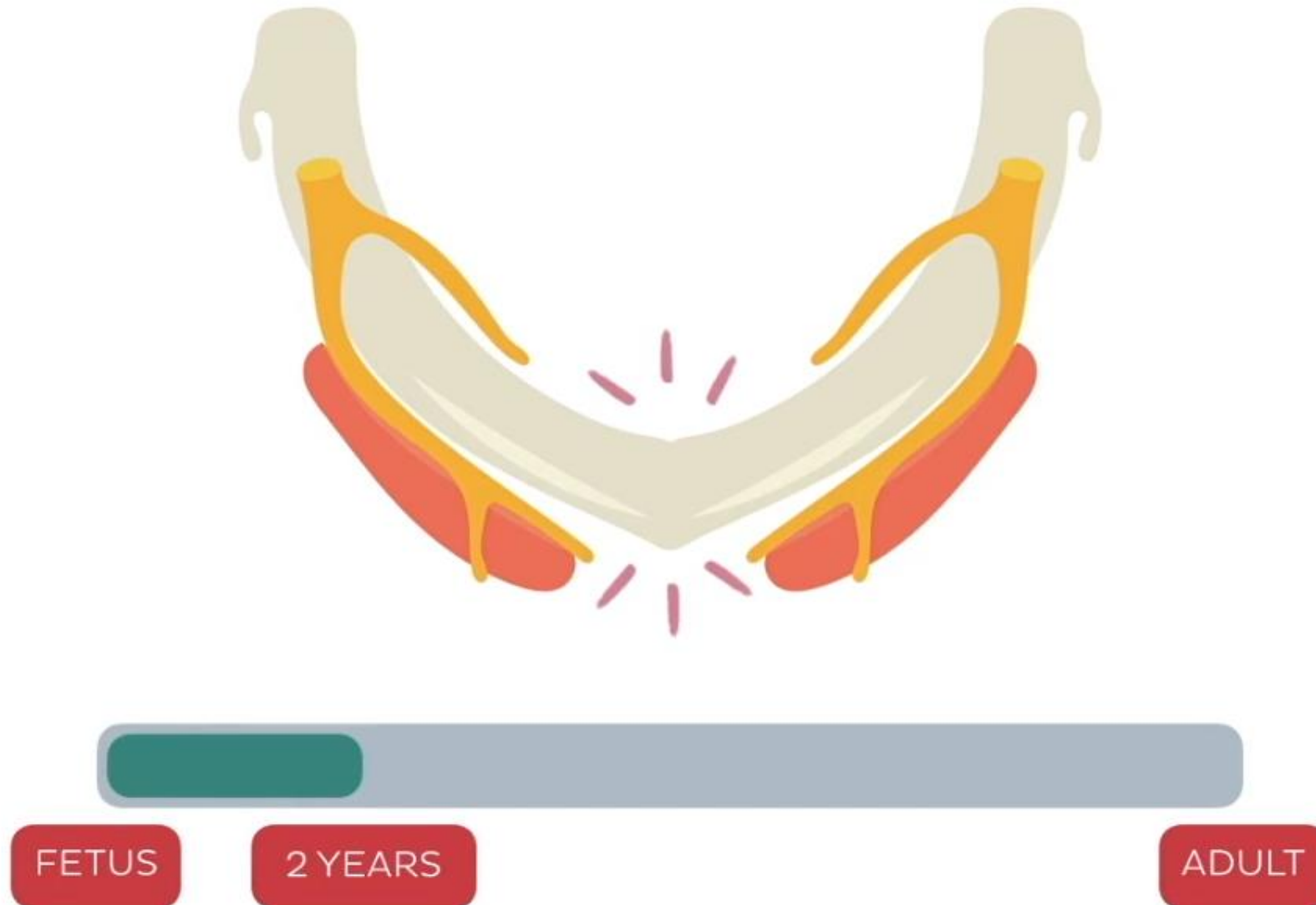


External surface



Internal surface

Development of the mandible



**Ridge at midline
(symphysis menti)**

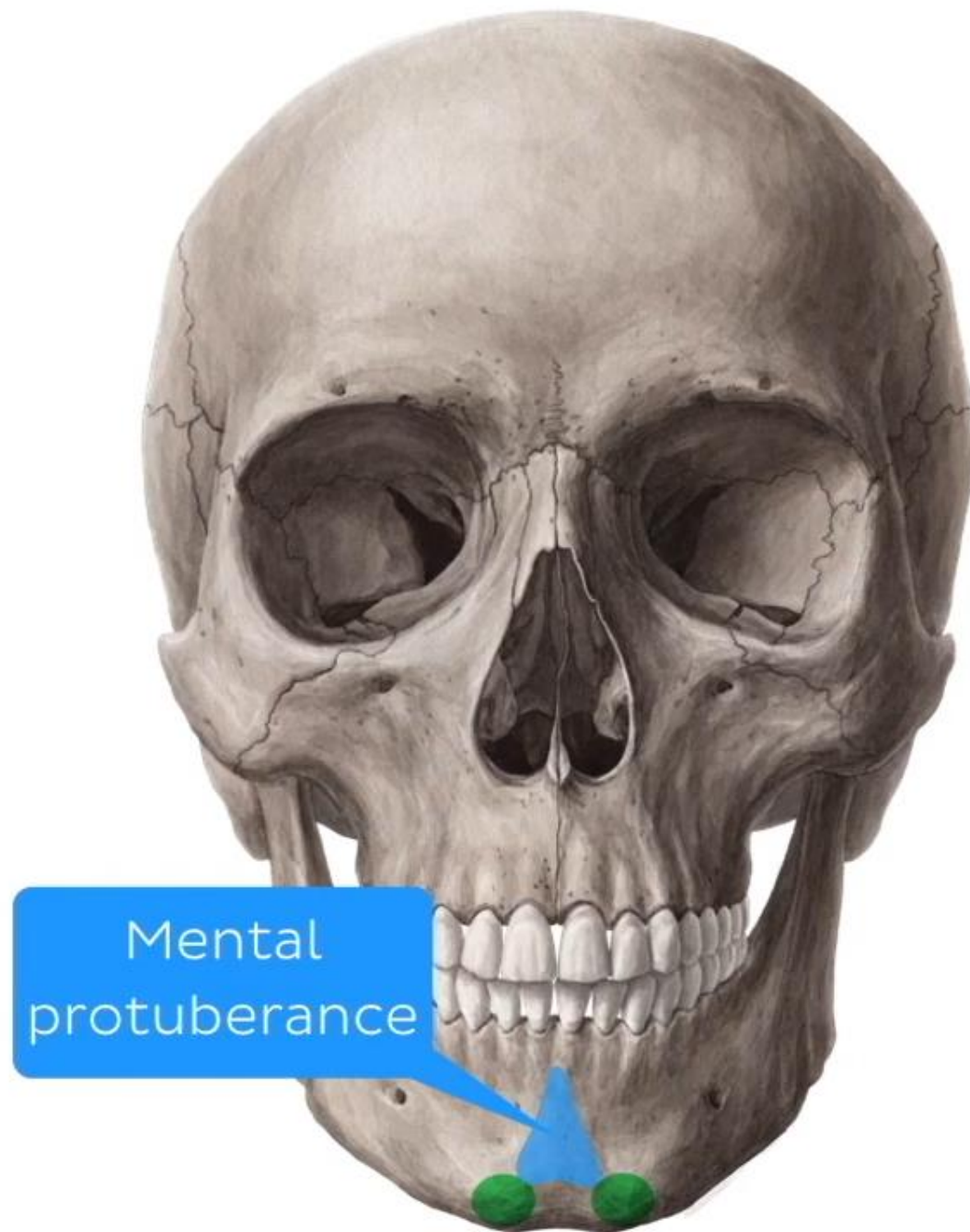


**Mental
protuberance**



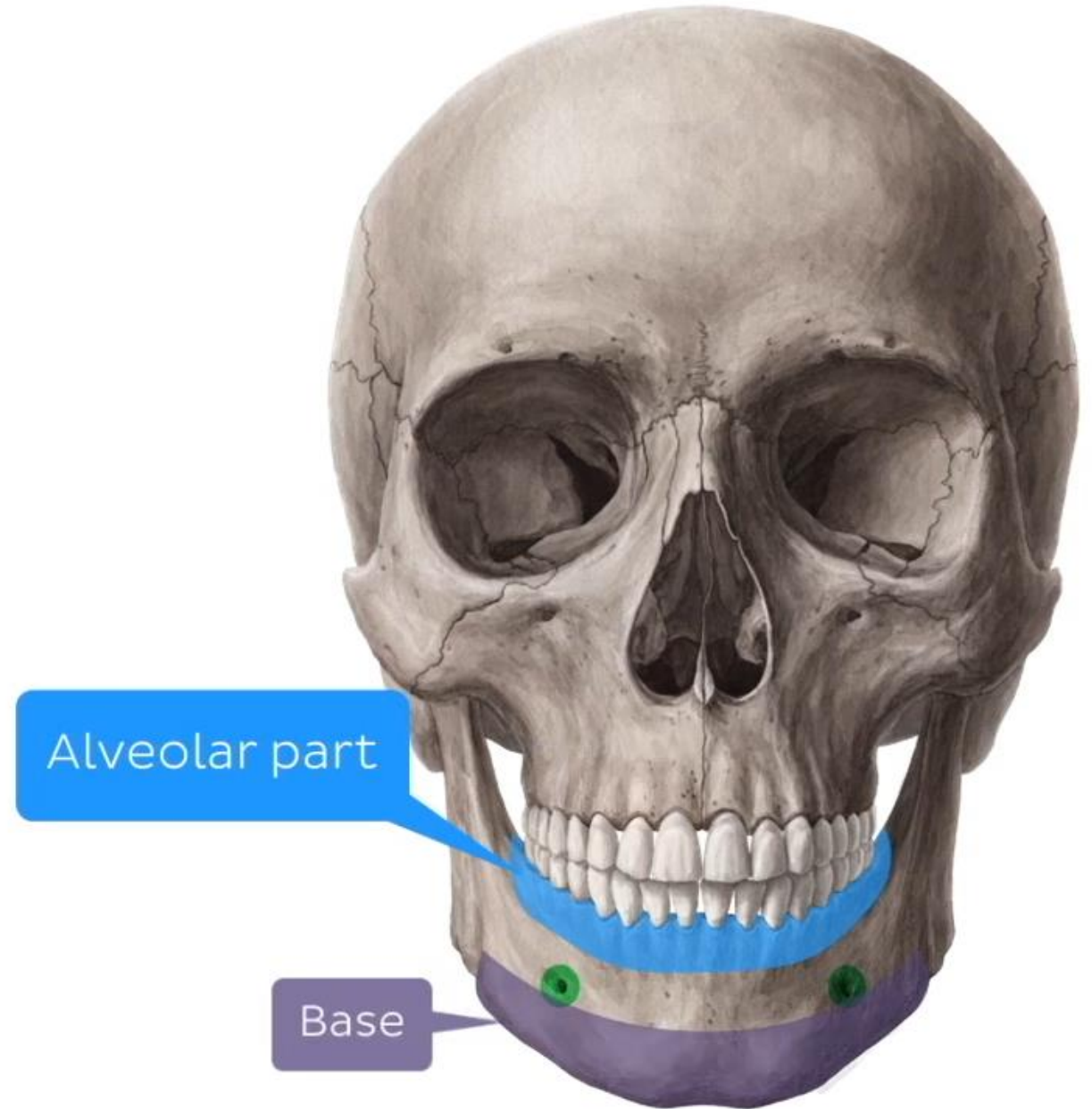
**Mental
tubercles**



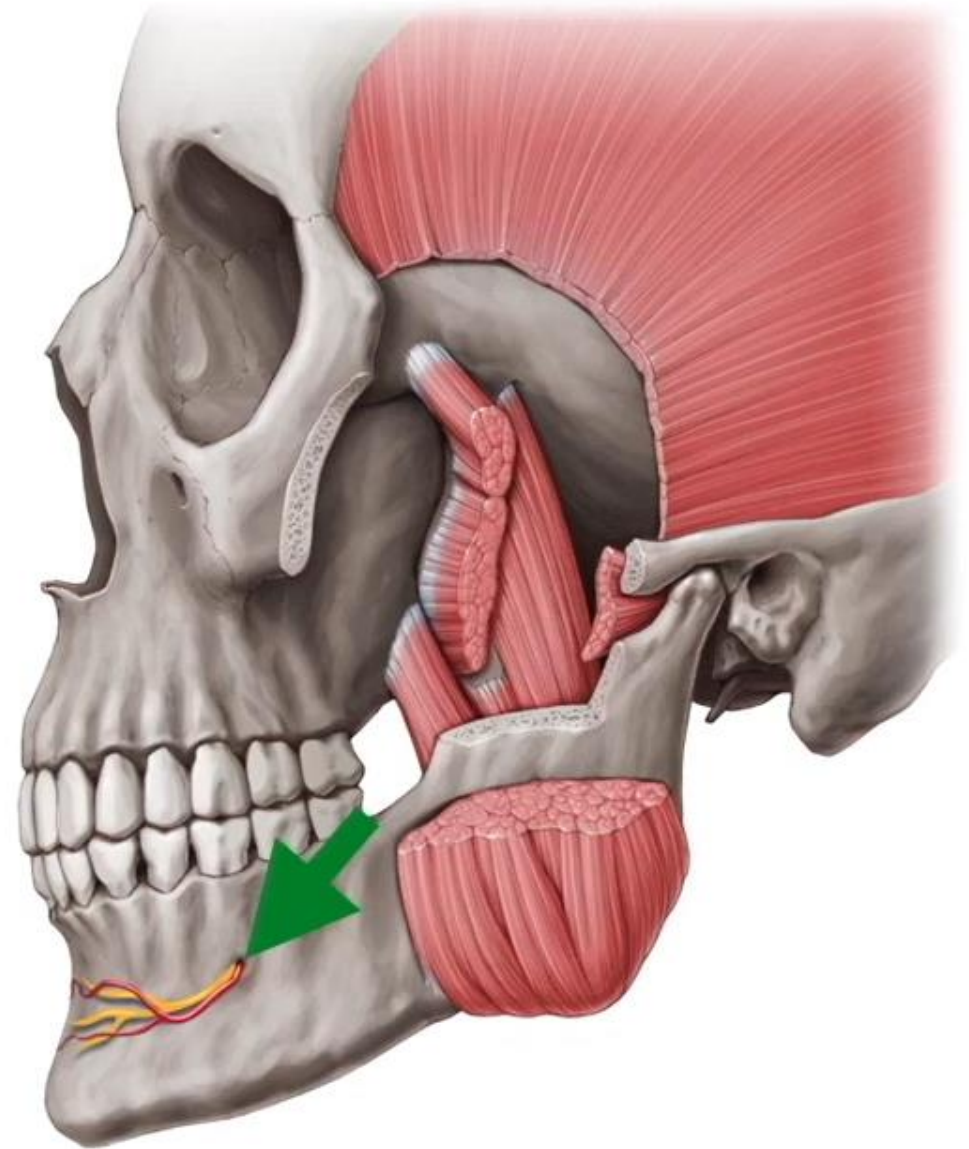
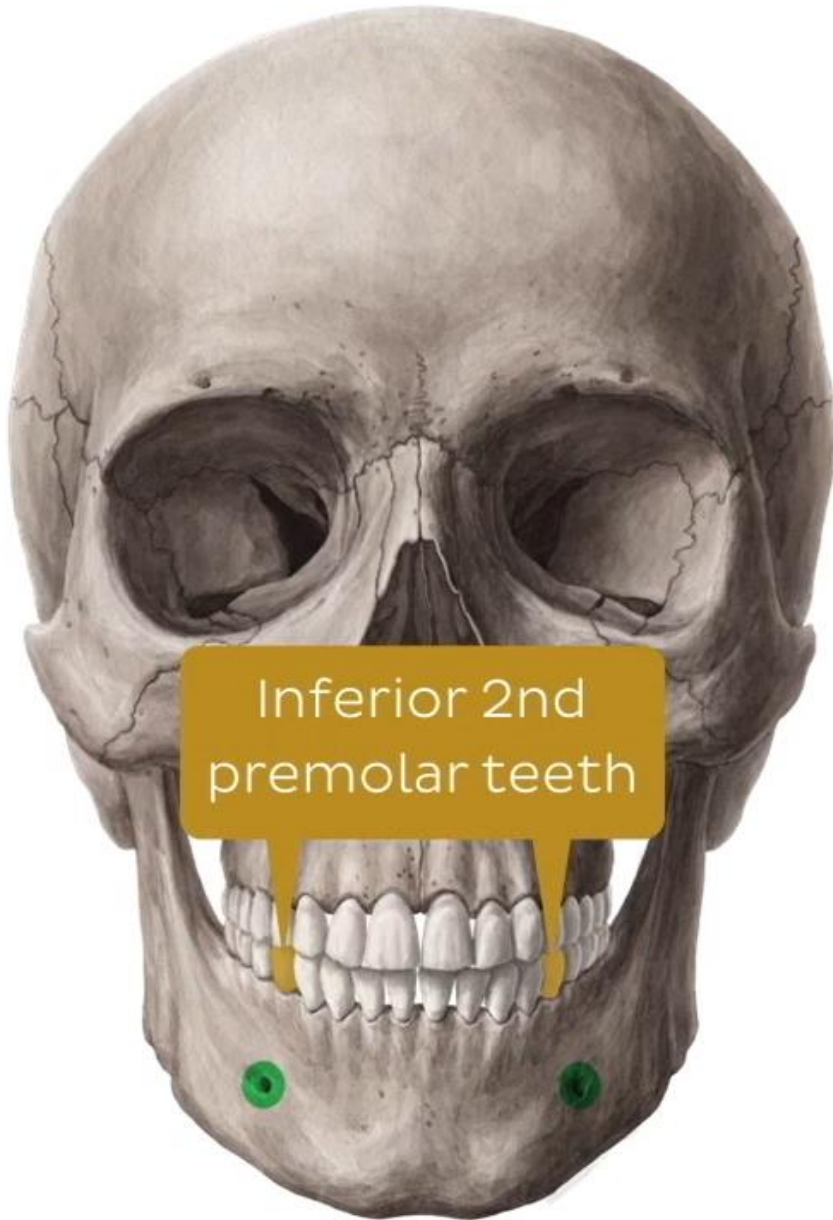


Mental
protuberance

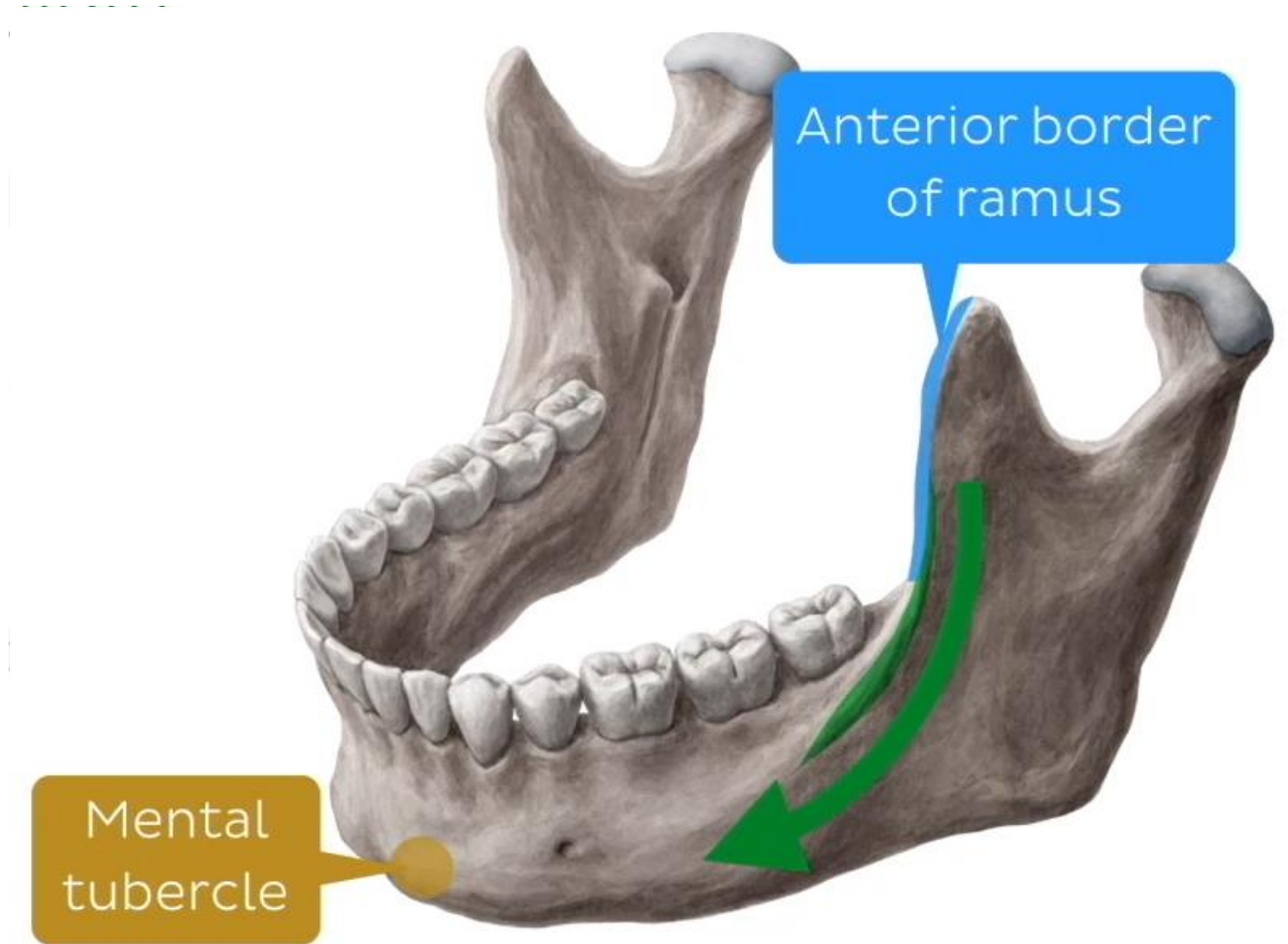
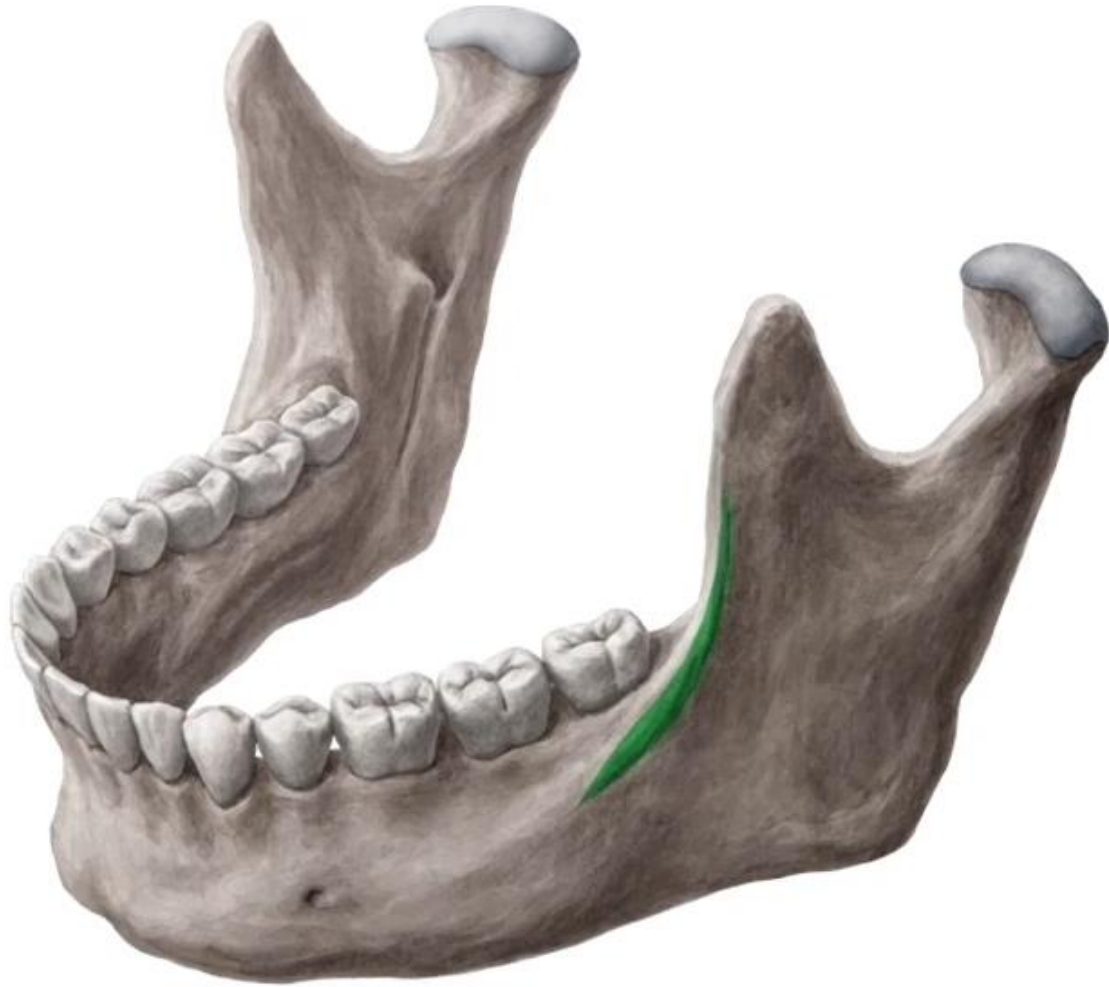
Mental foramen



Mental foramen



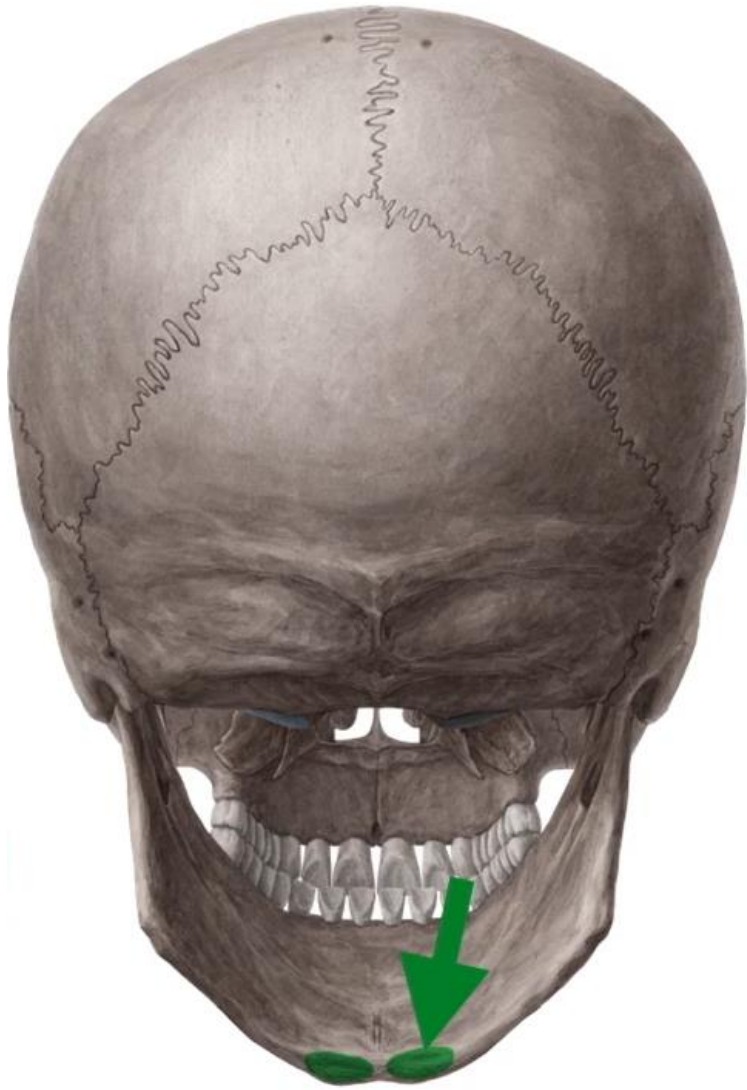
External oblique ridge



Internal body of the mandible



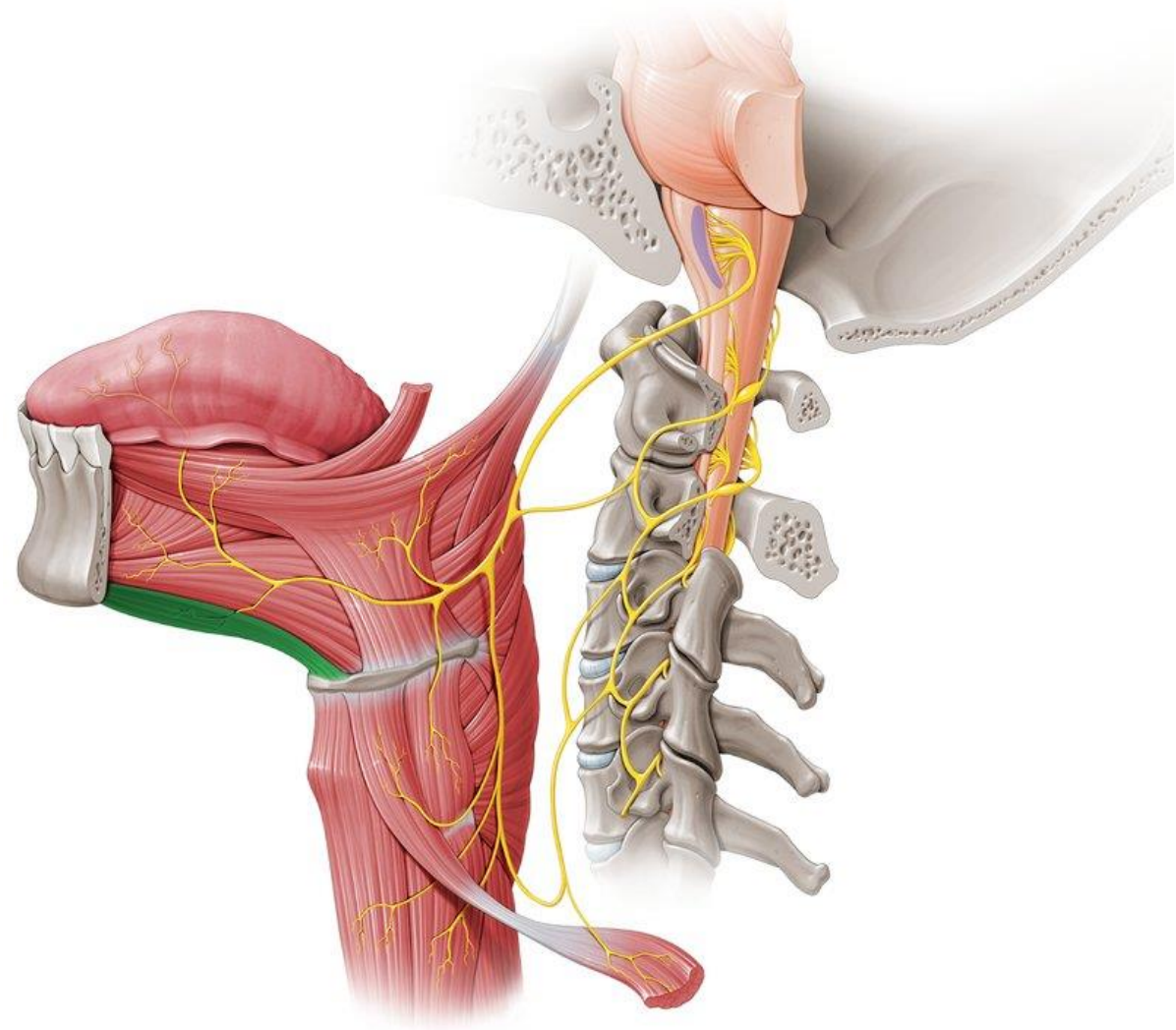
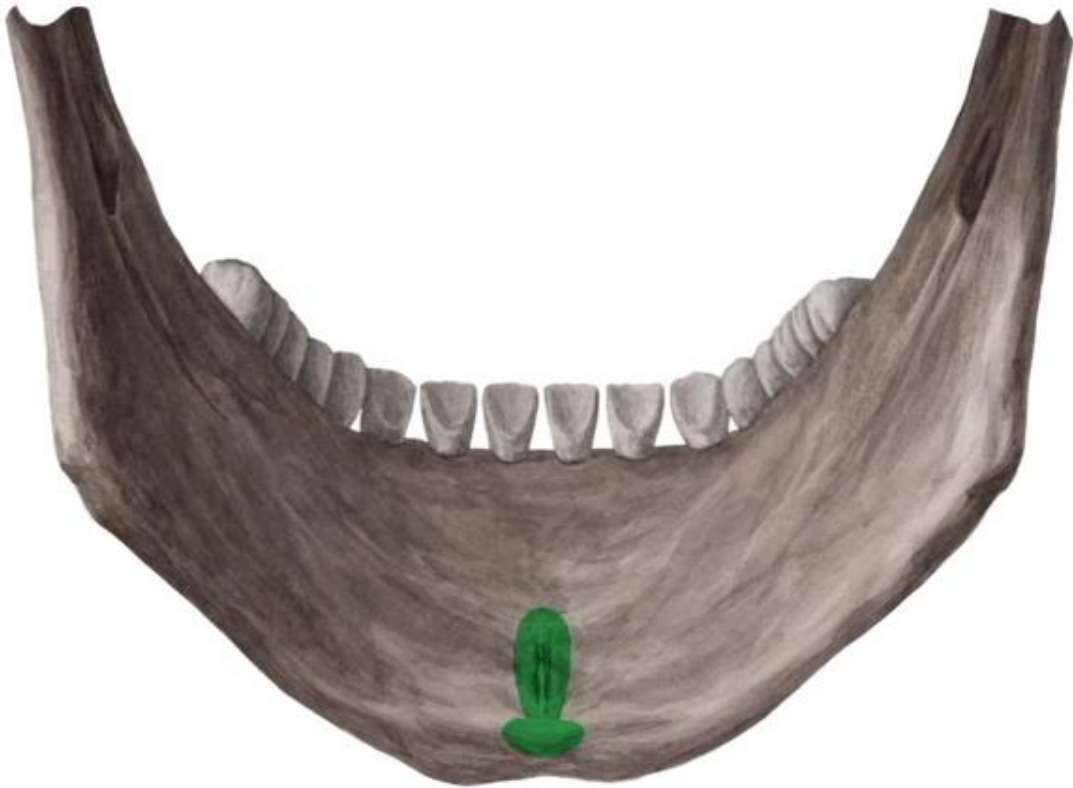
Digastric fossae



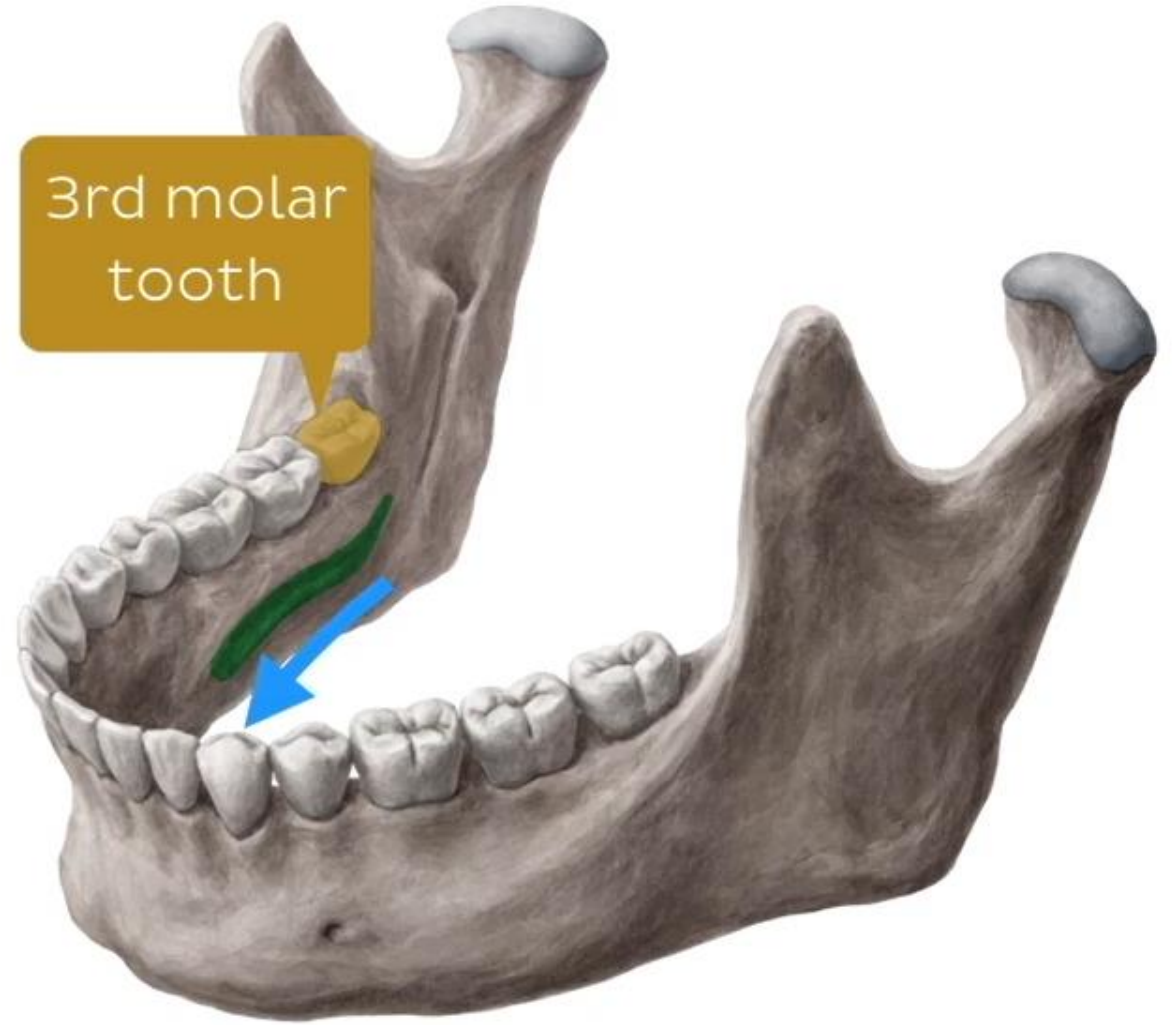
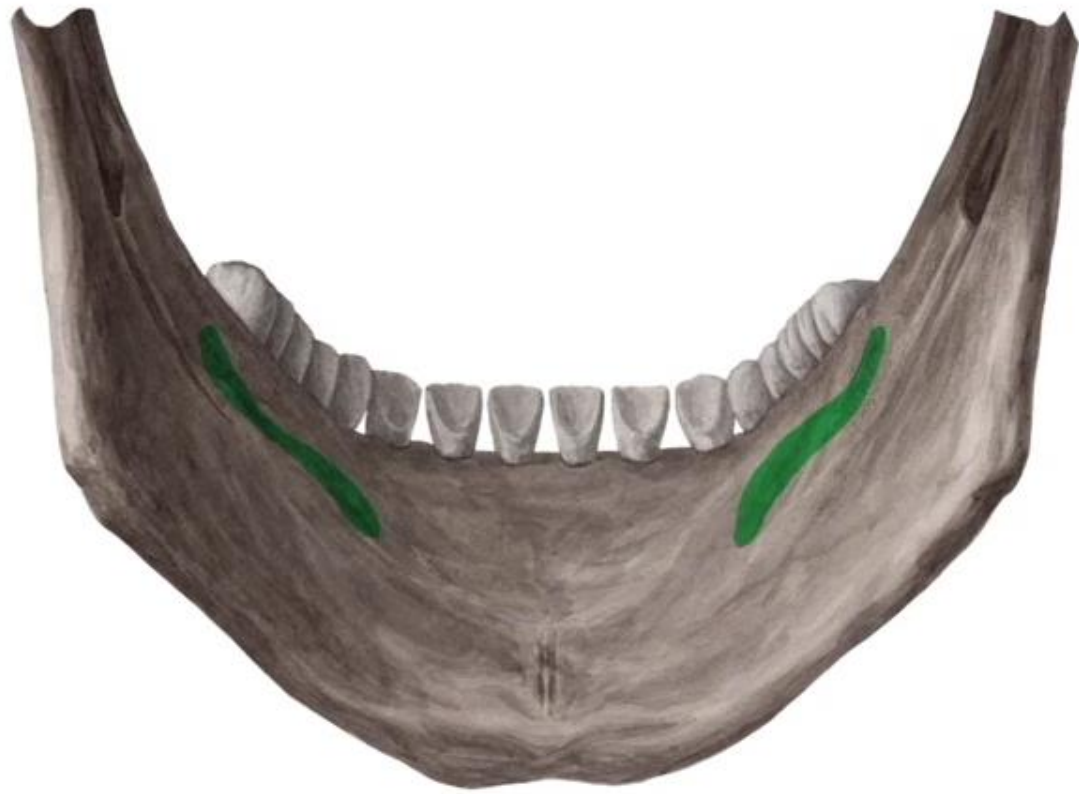
Anterior bellies of
digastric muscles



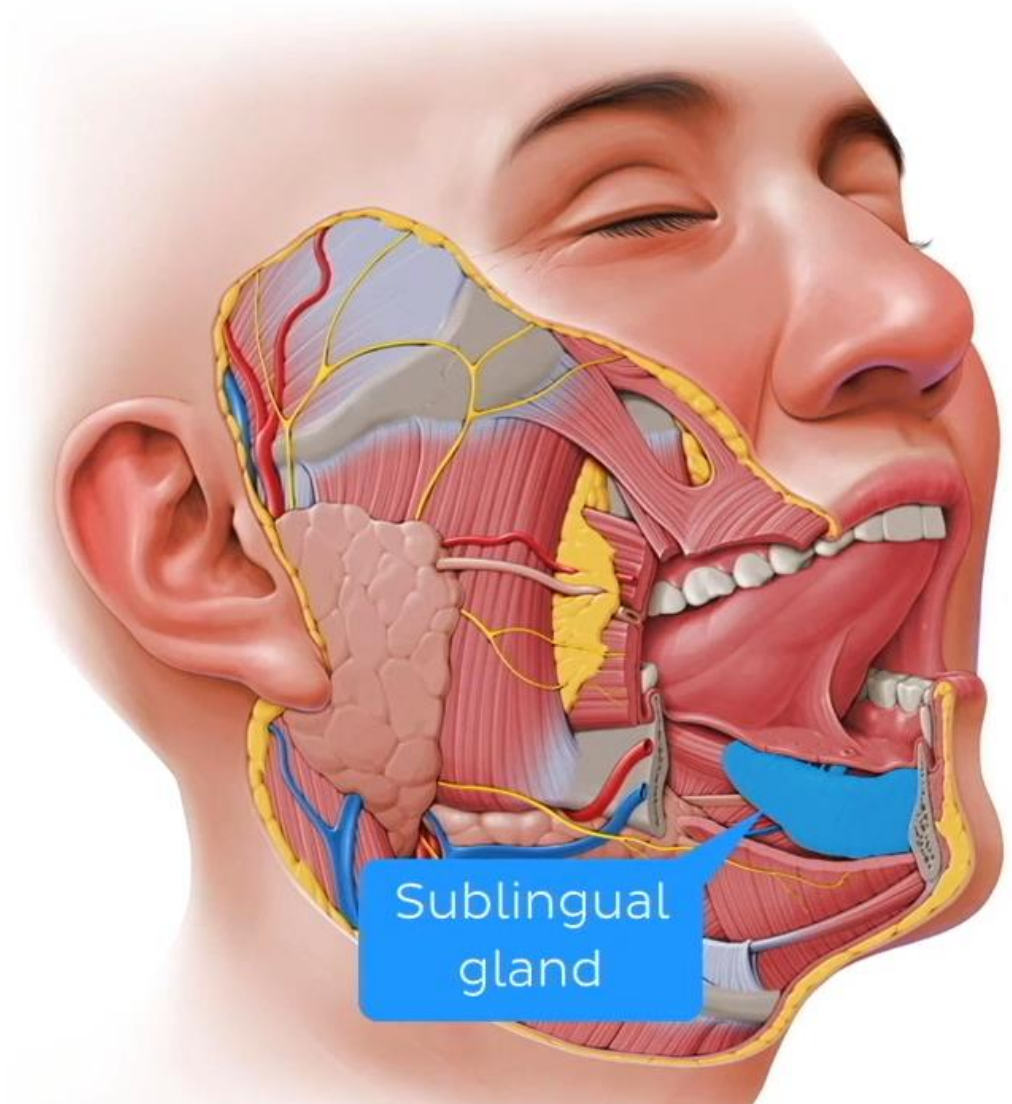
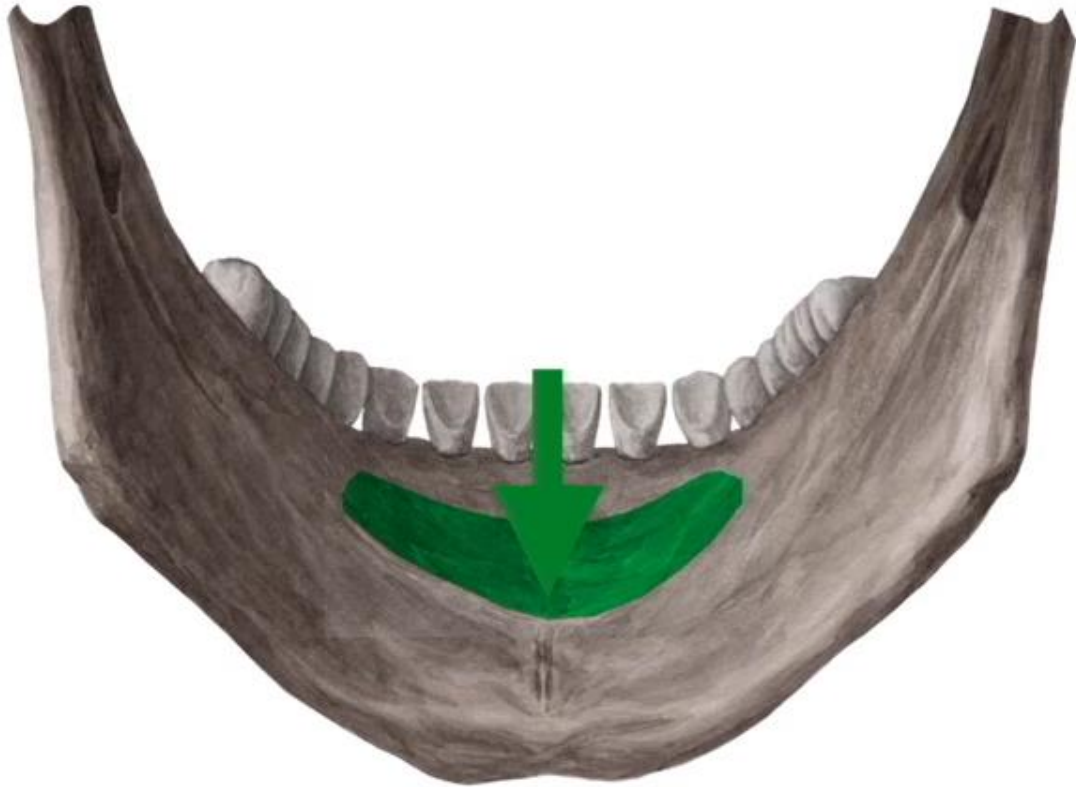
Genial tubercles



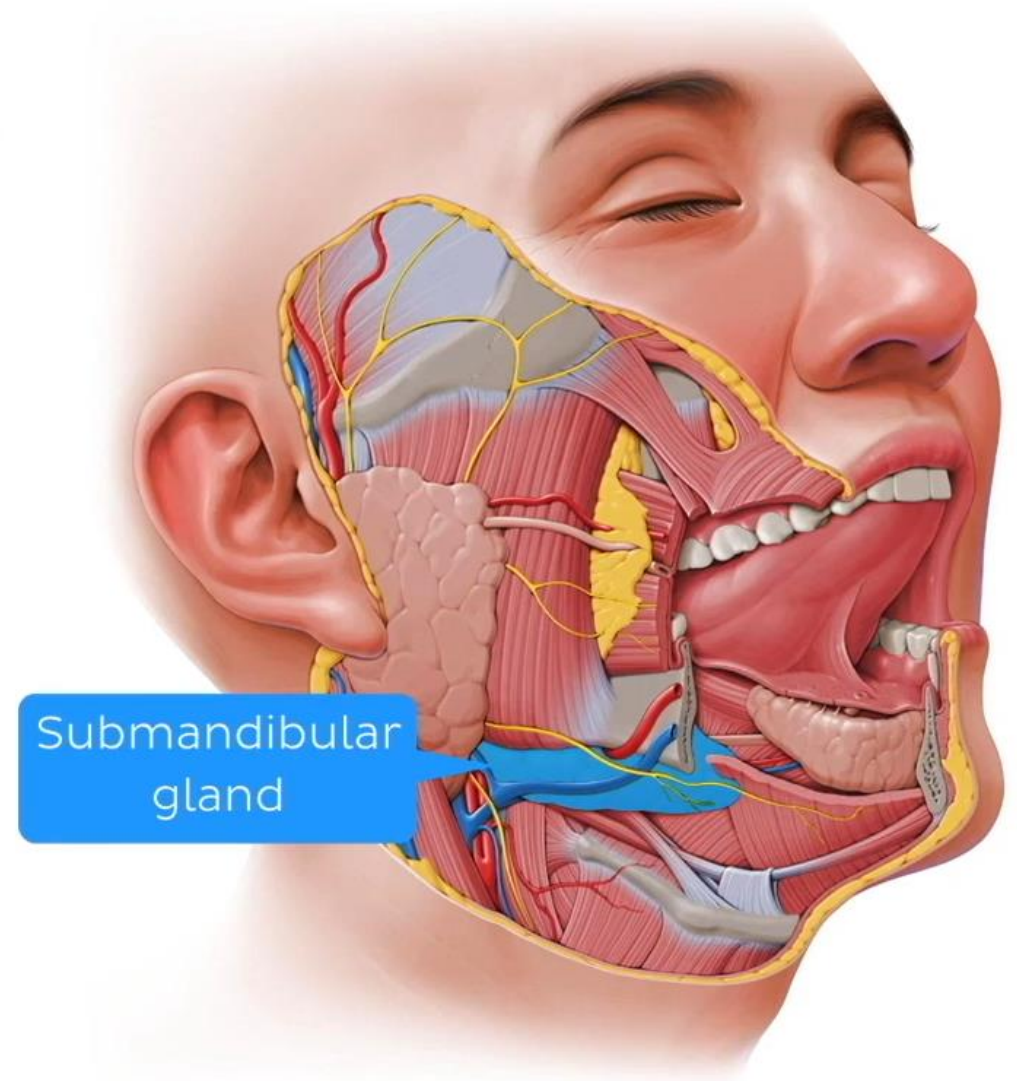
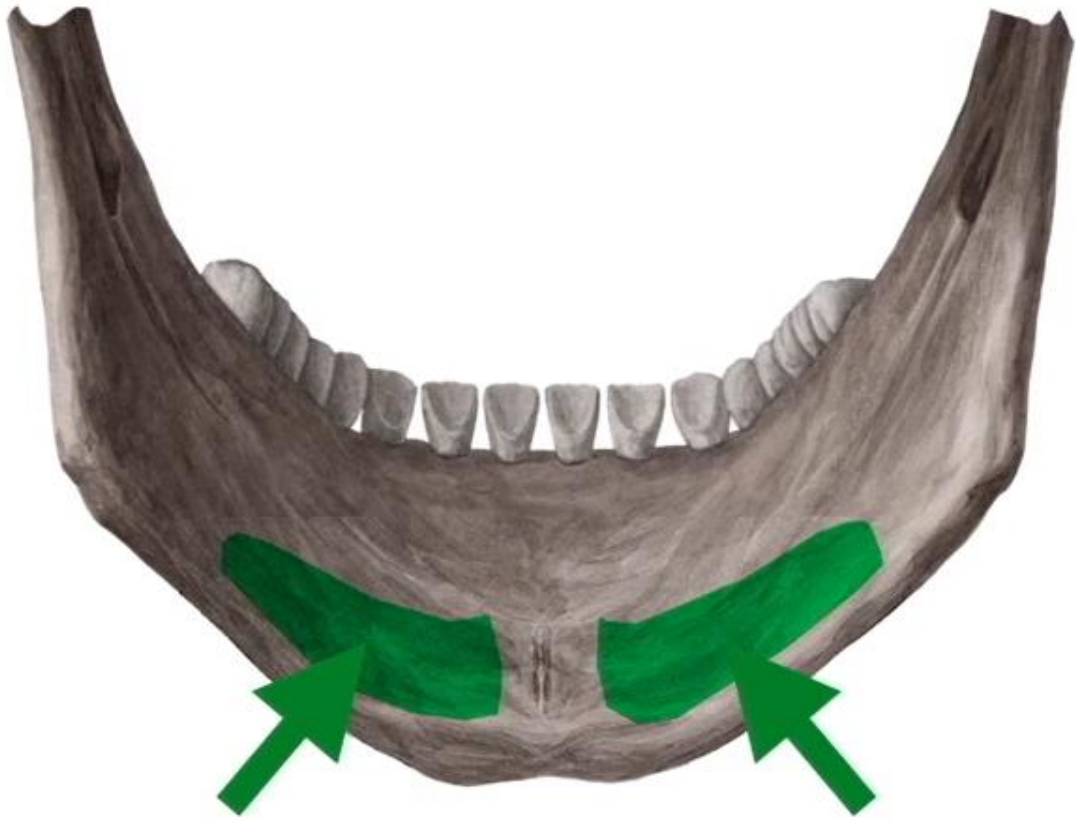
Mylohyoid line



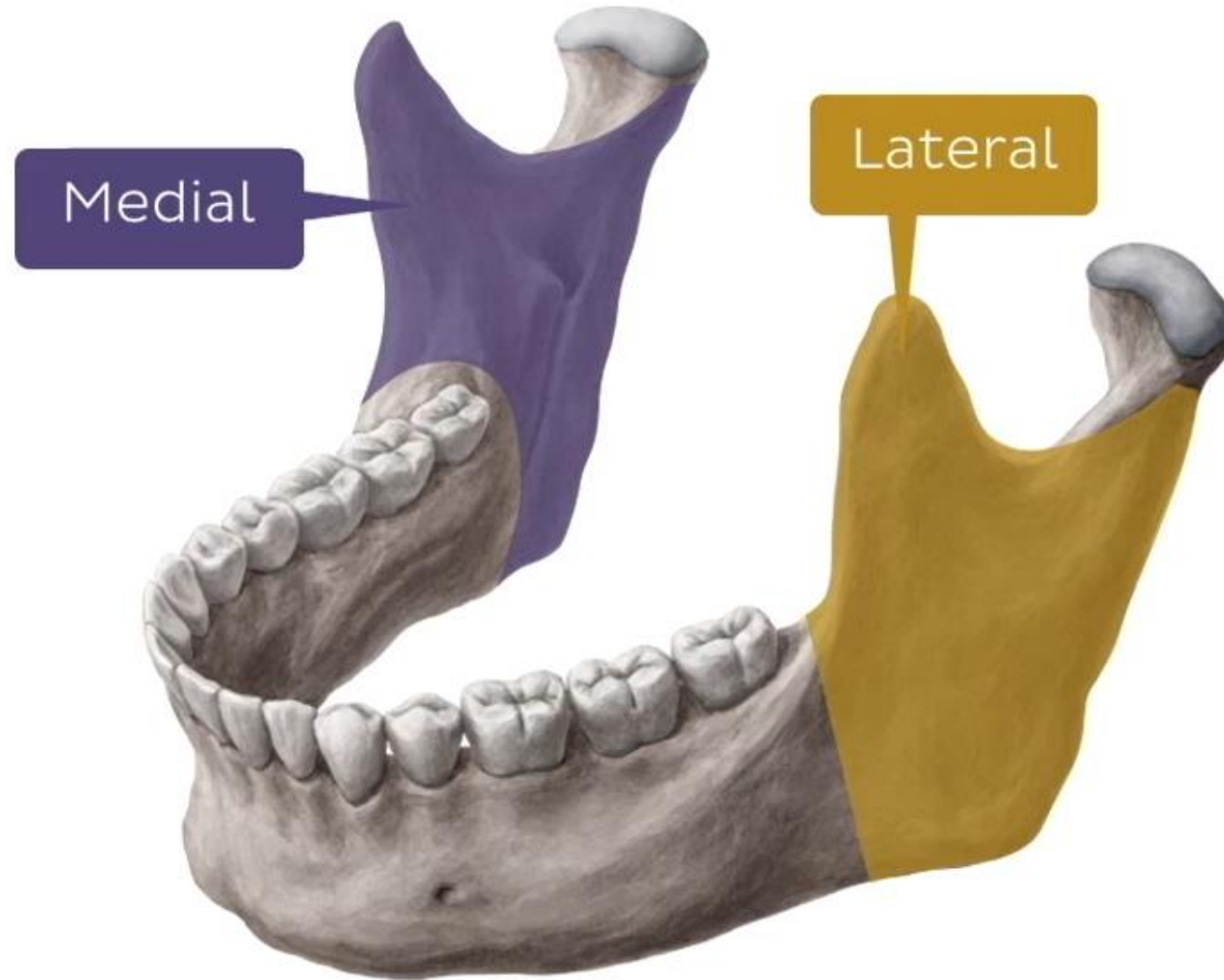
Sublingual fossa



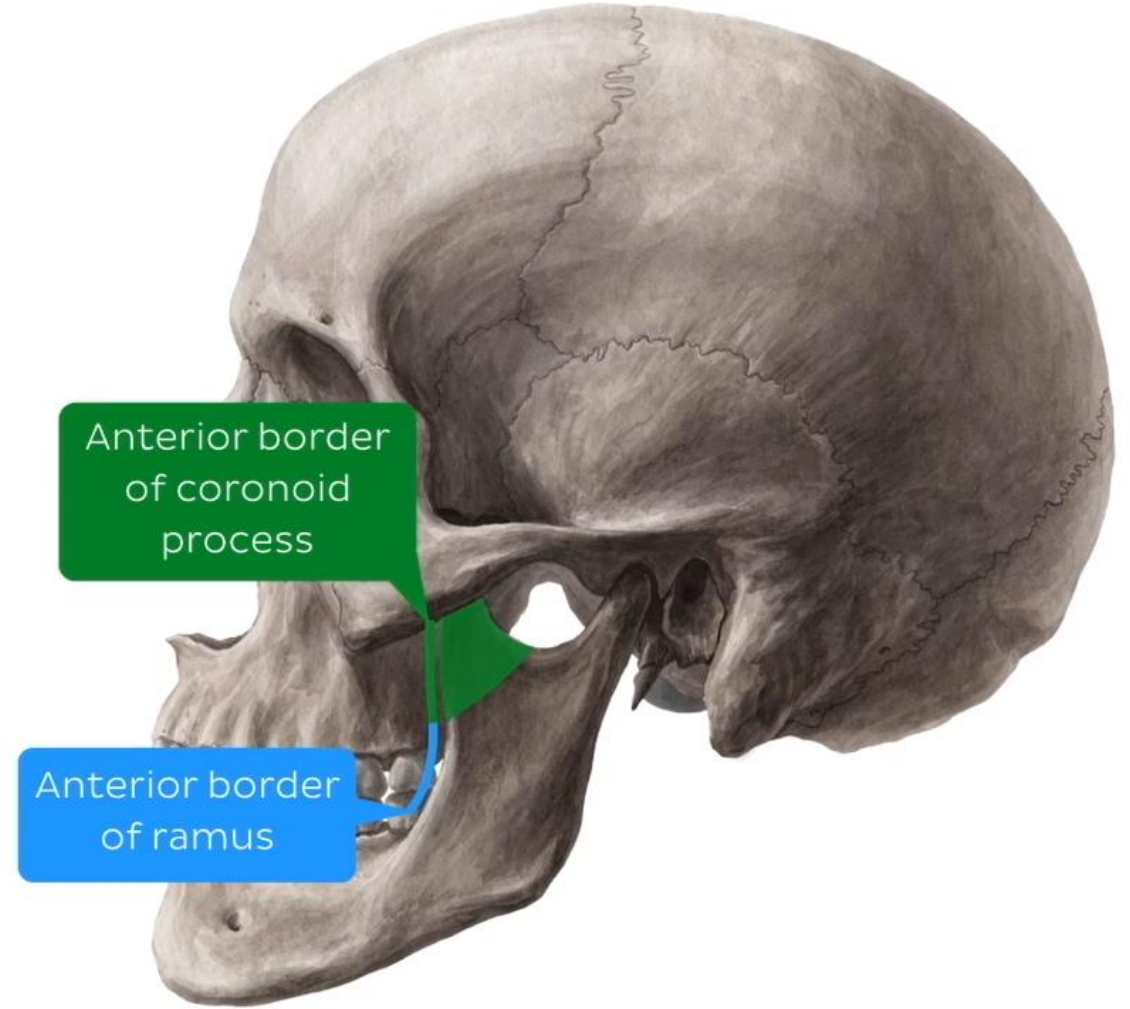
Submandibular fossa



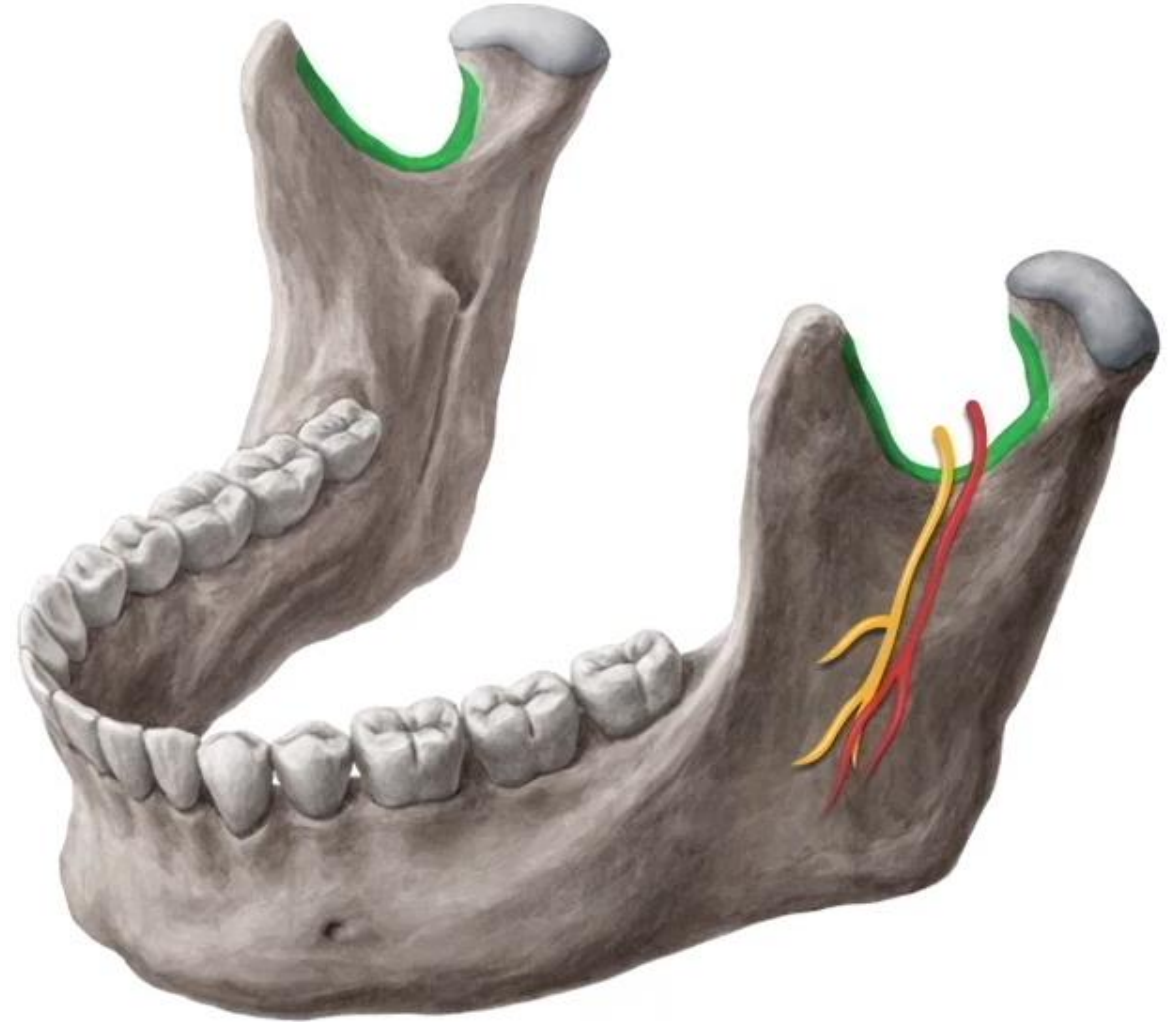
Ramus of the mandible



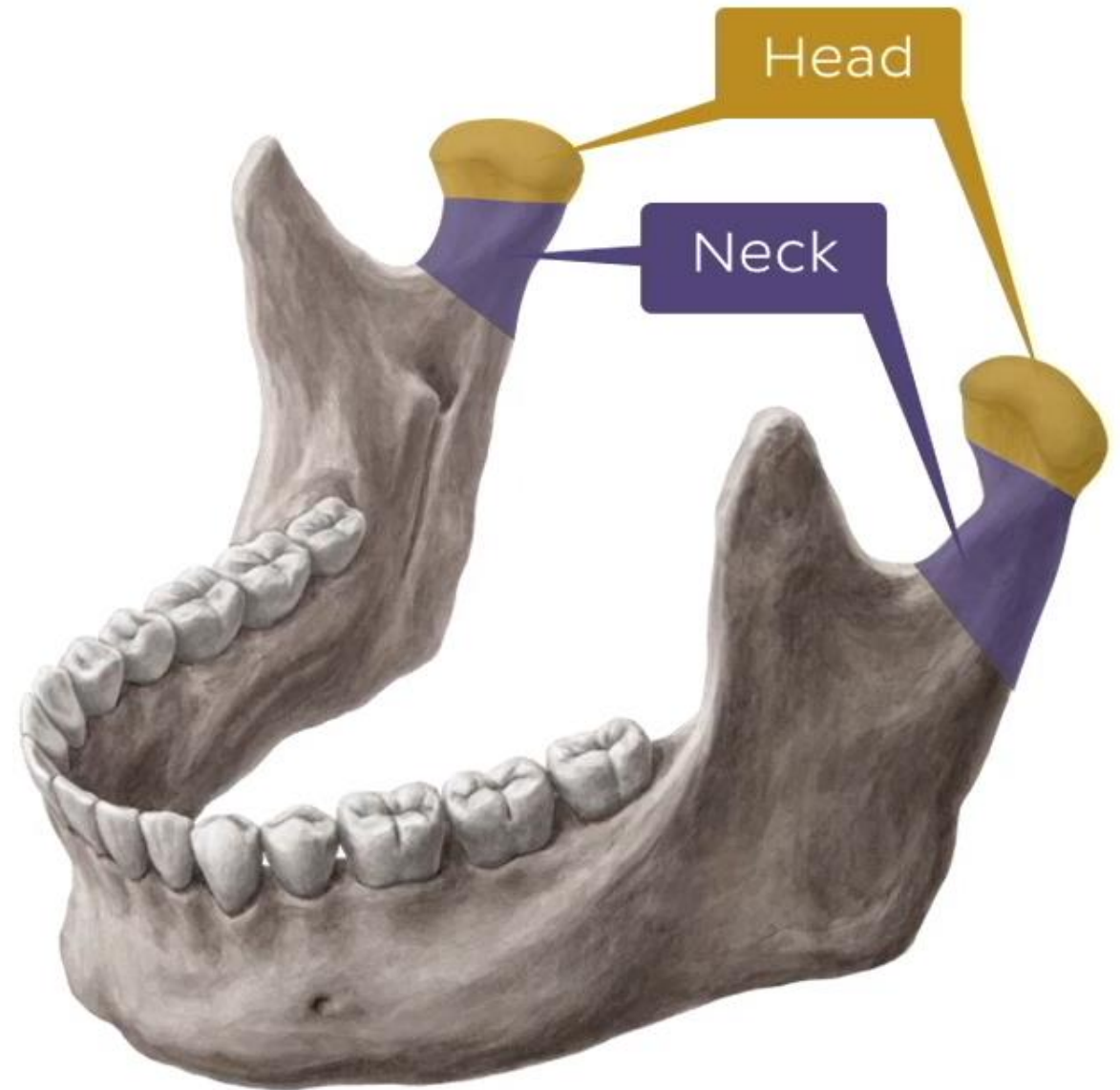
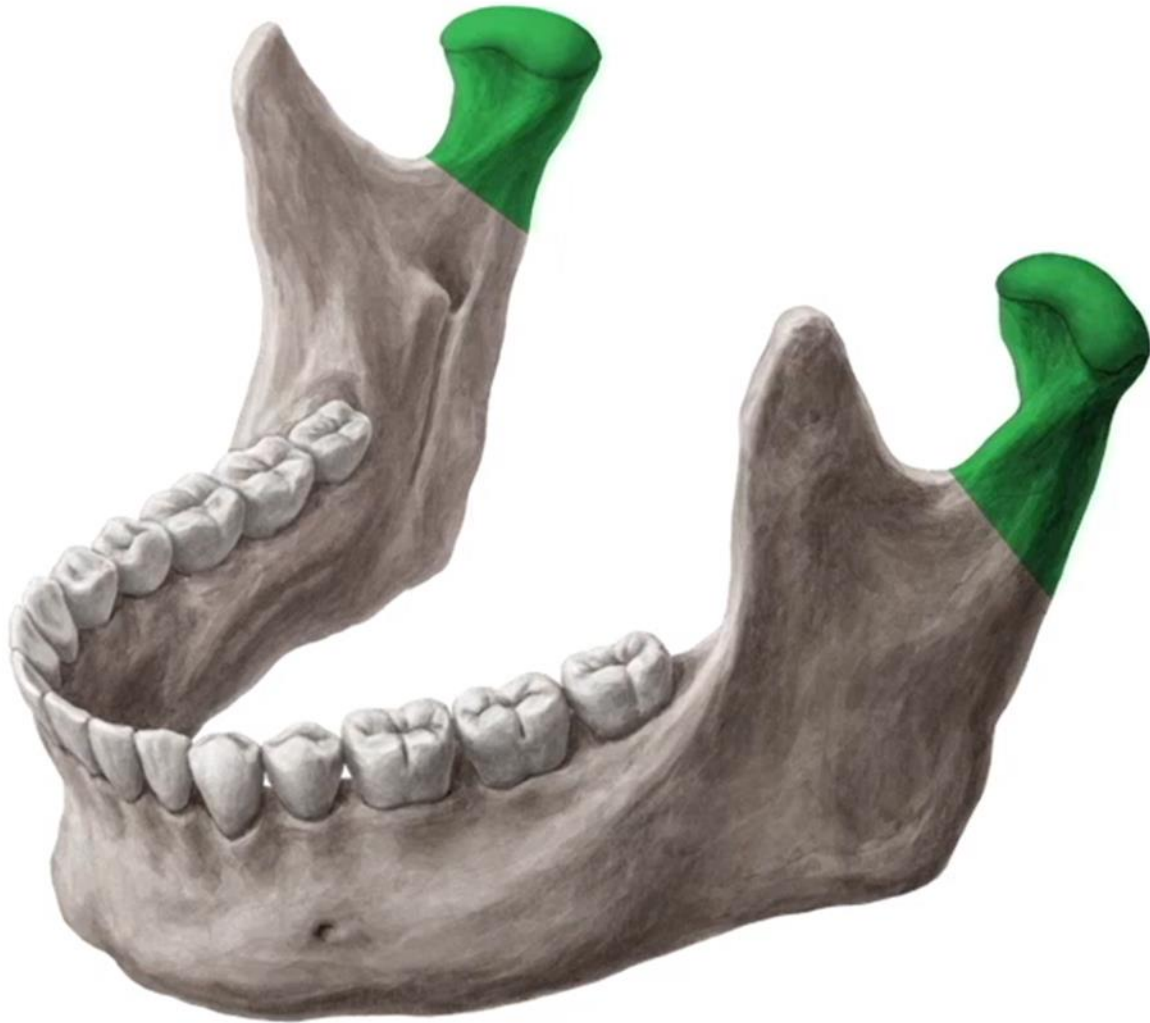
Coronoid process



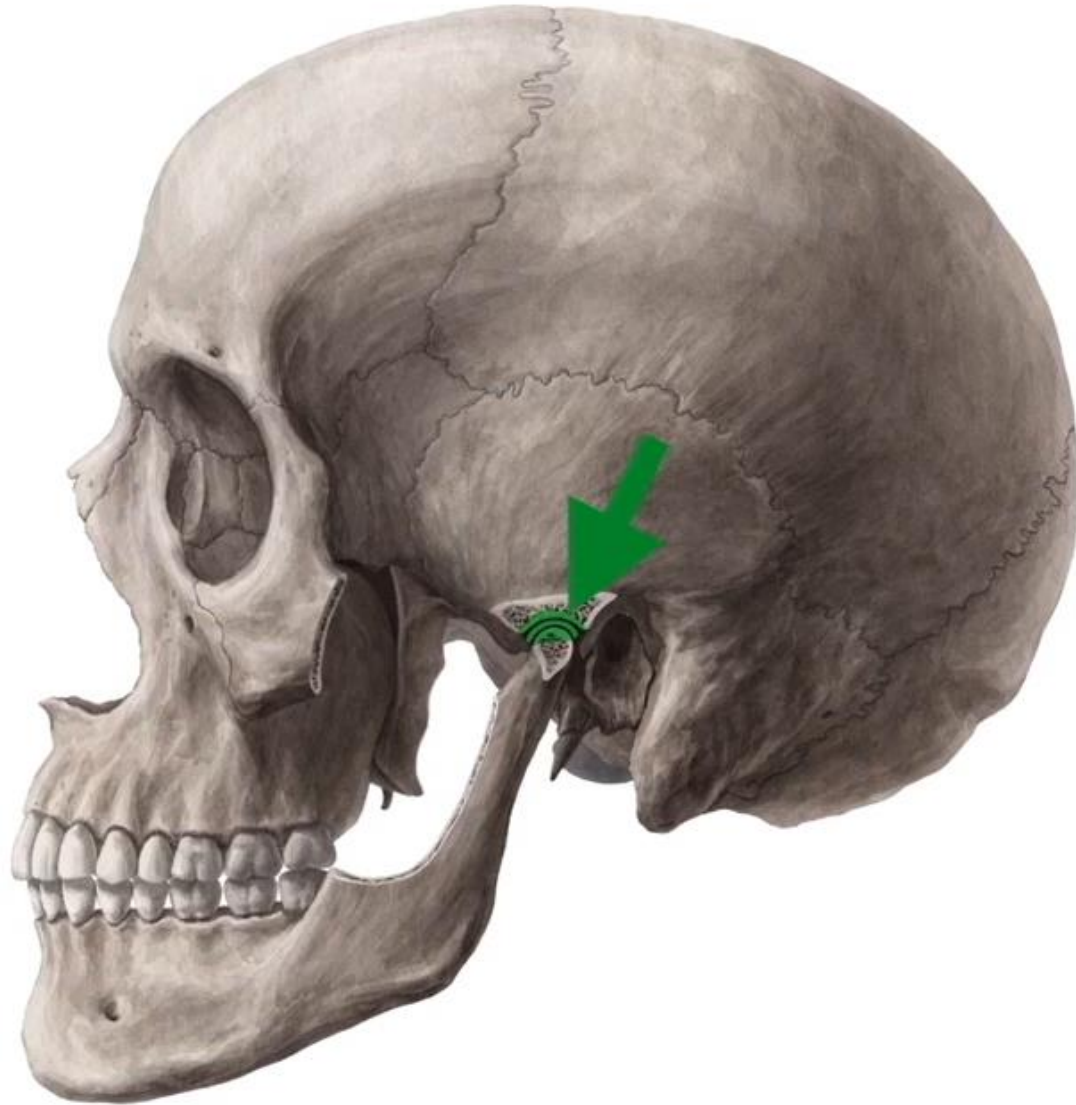
Mandibular/Sigmoid notch



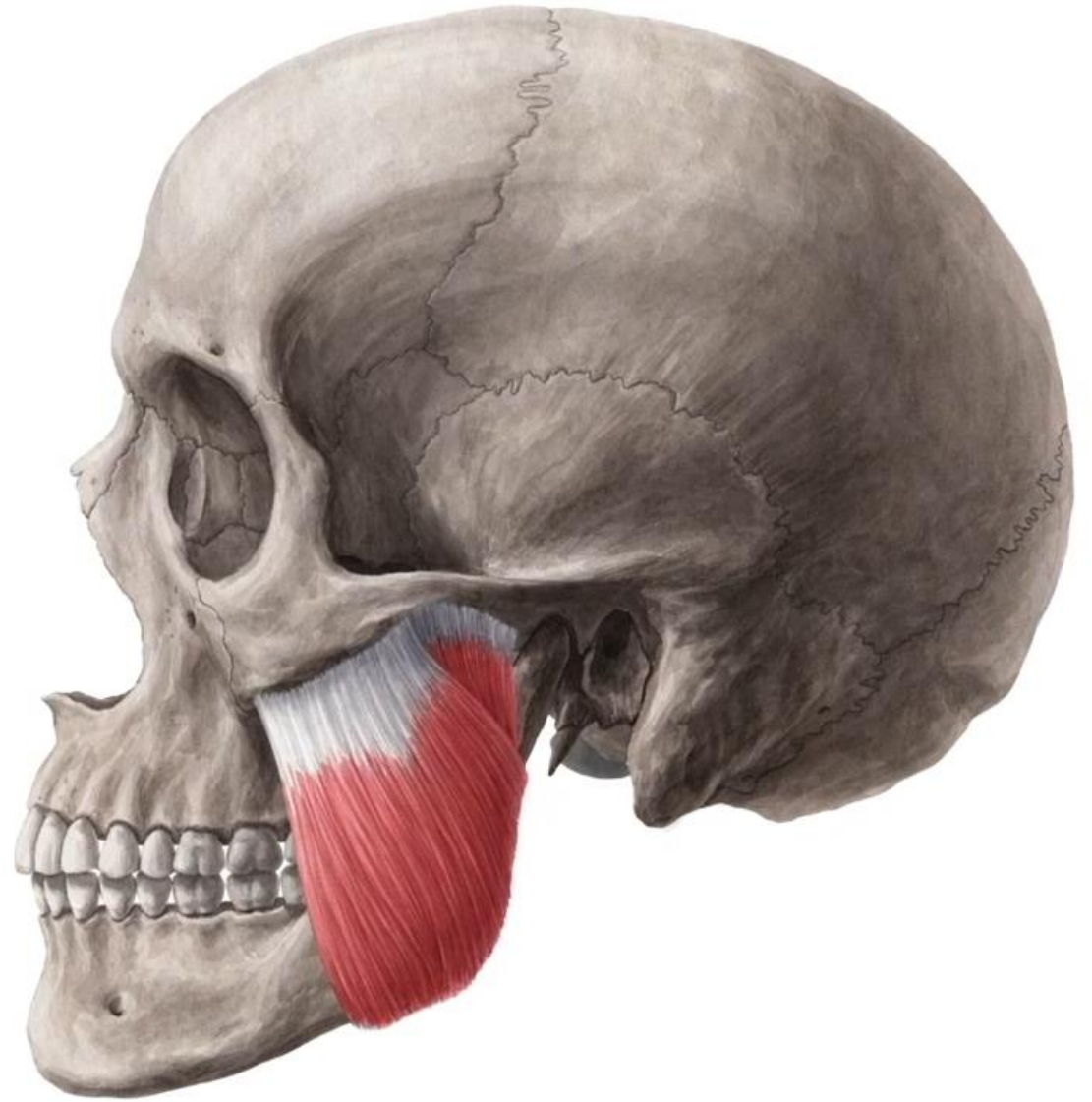
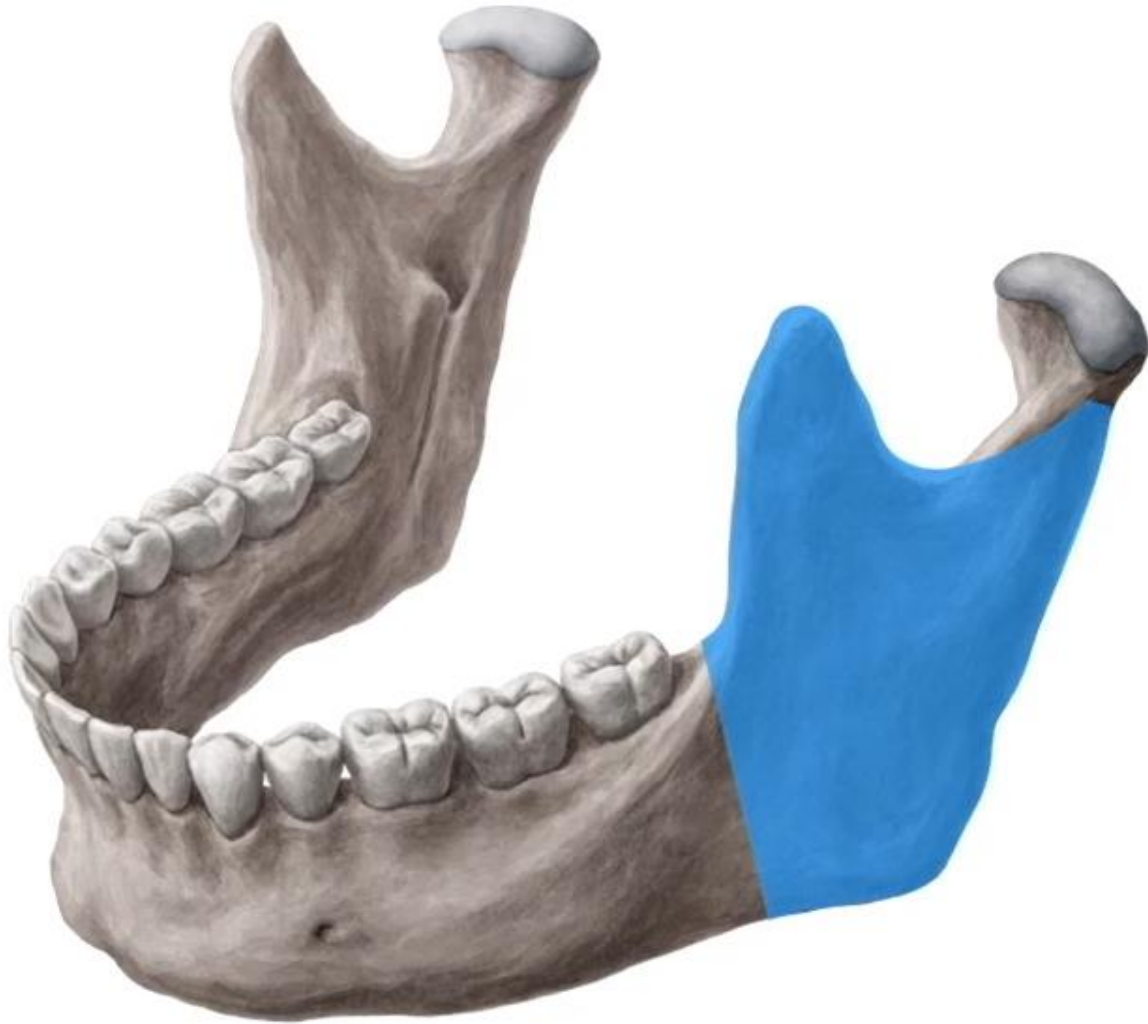
Condylar process



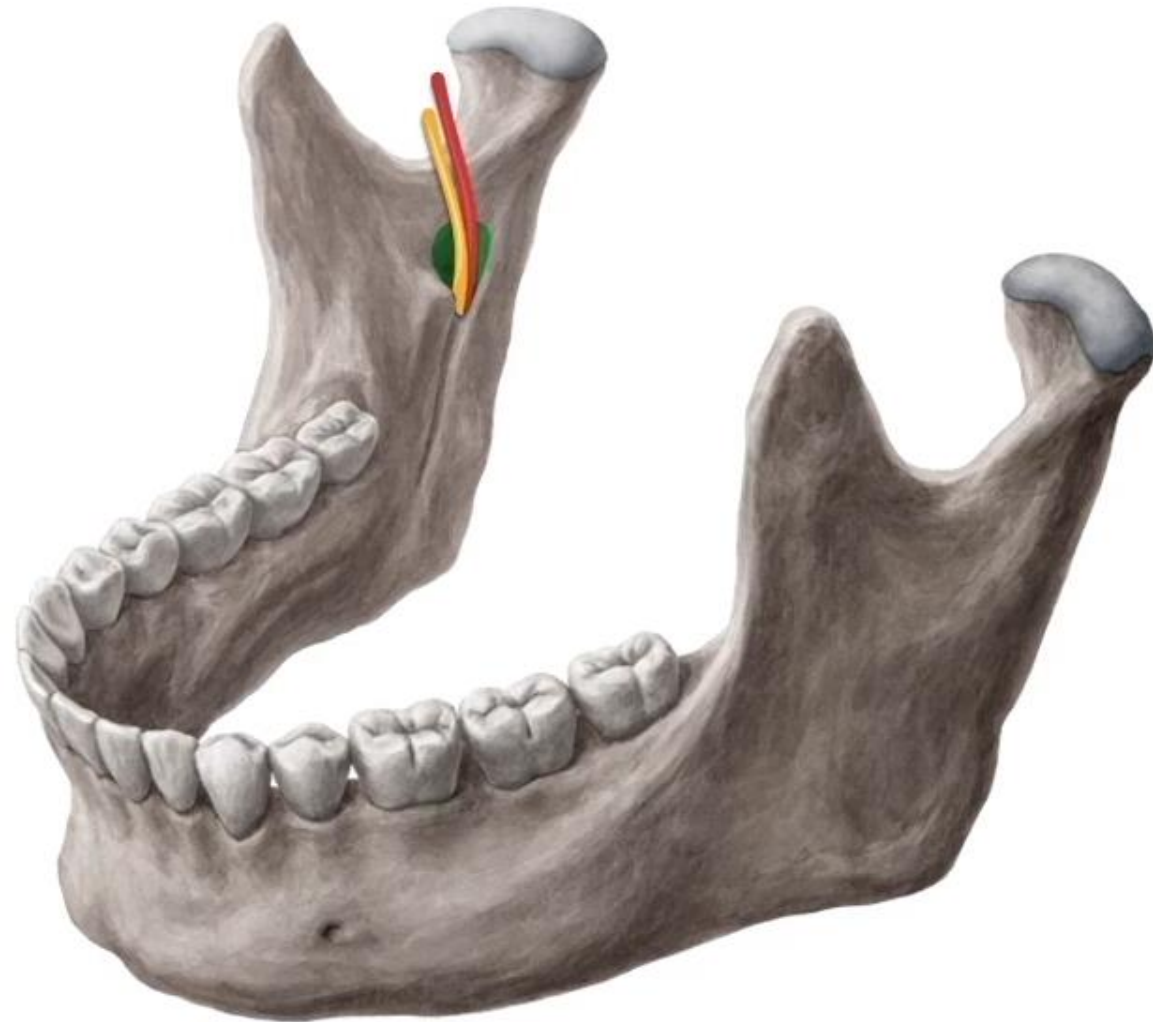
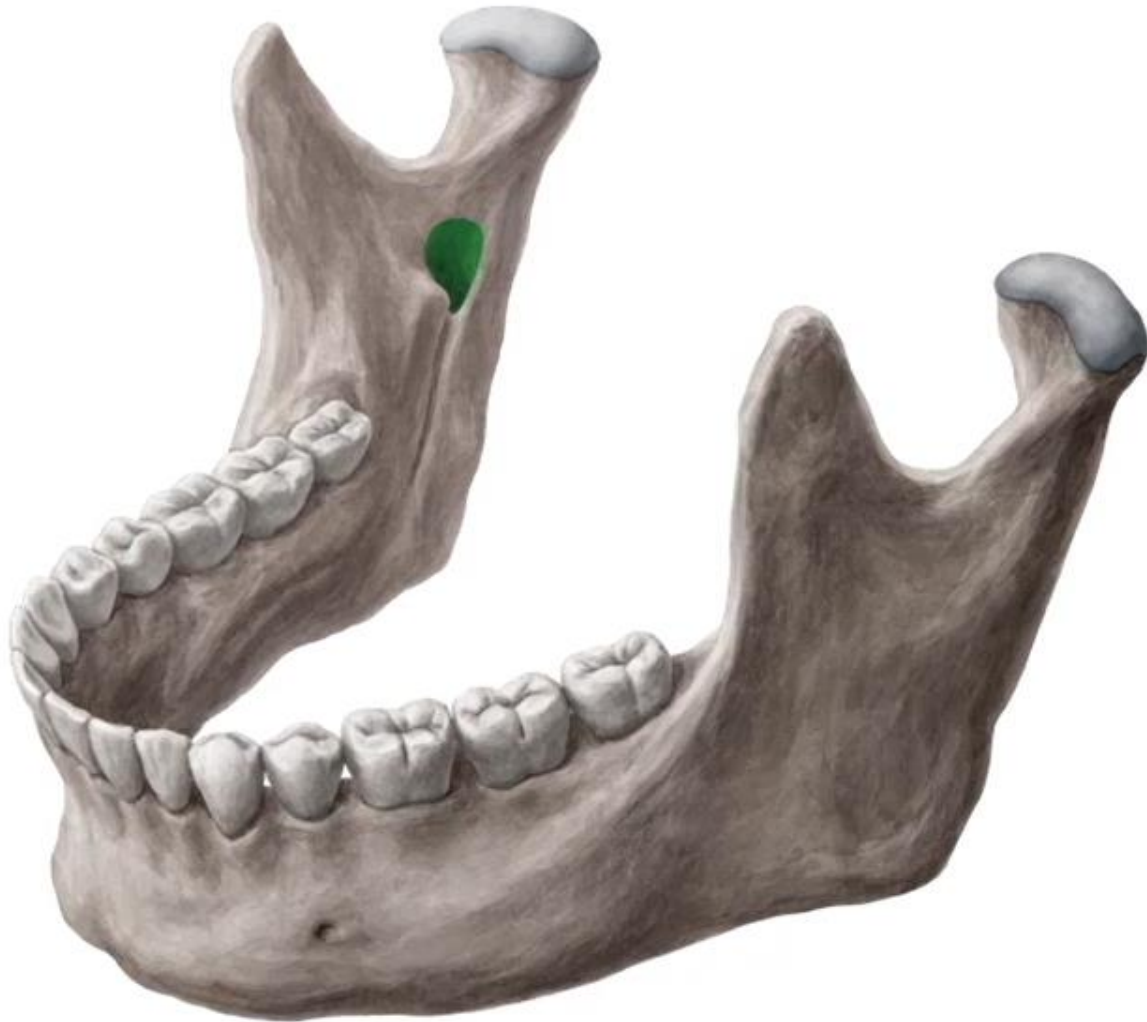
Temporomandibular joint



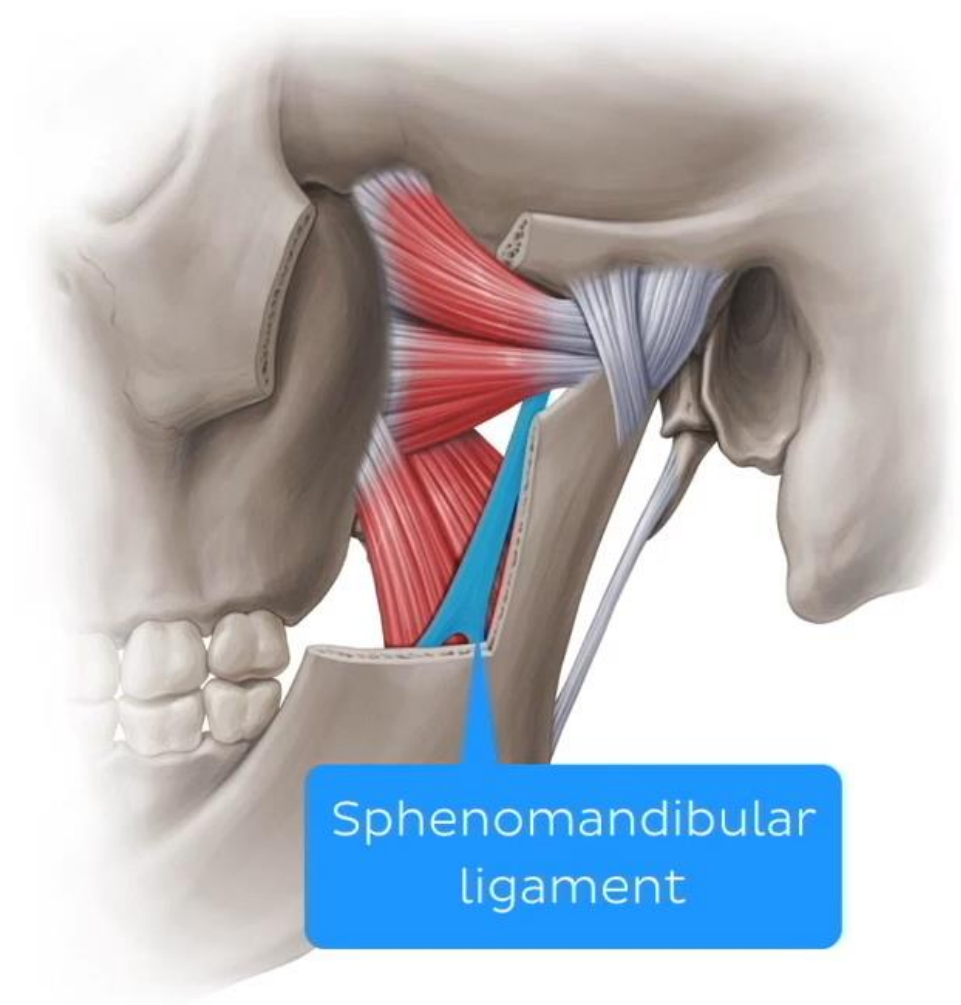
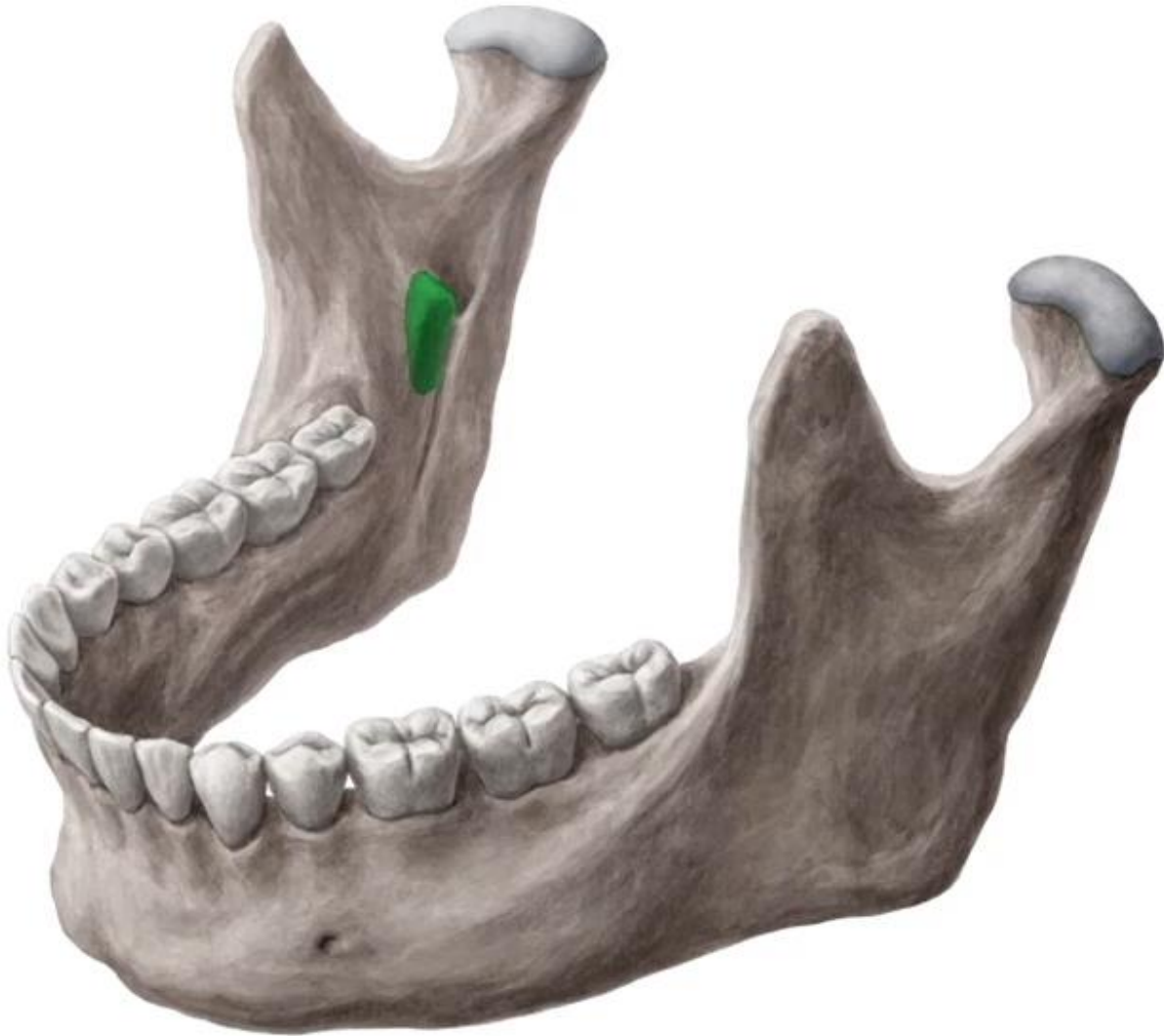
Lateral surface of the ramus



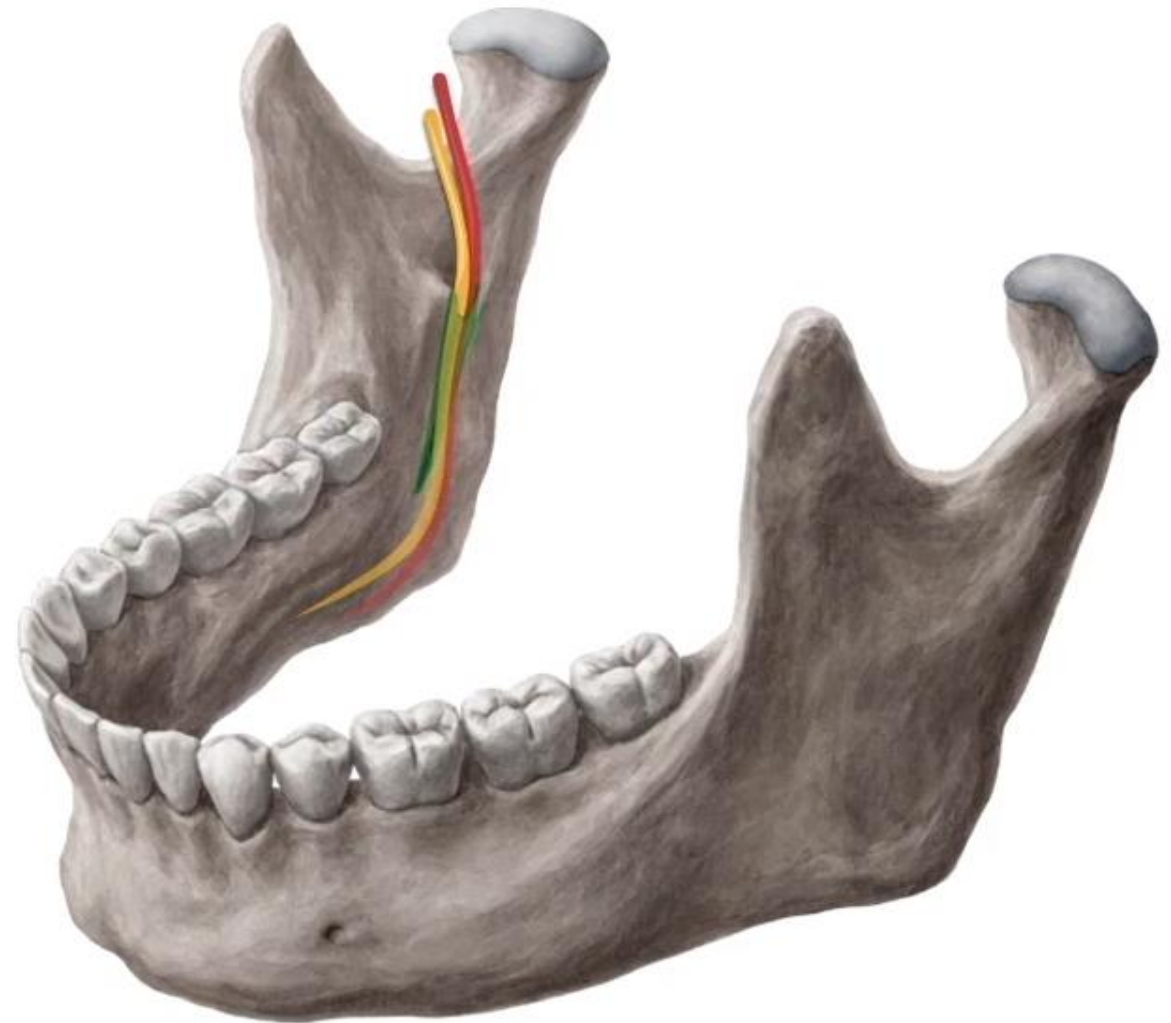
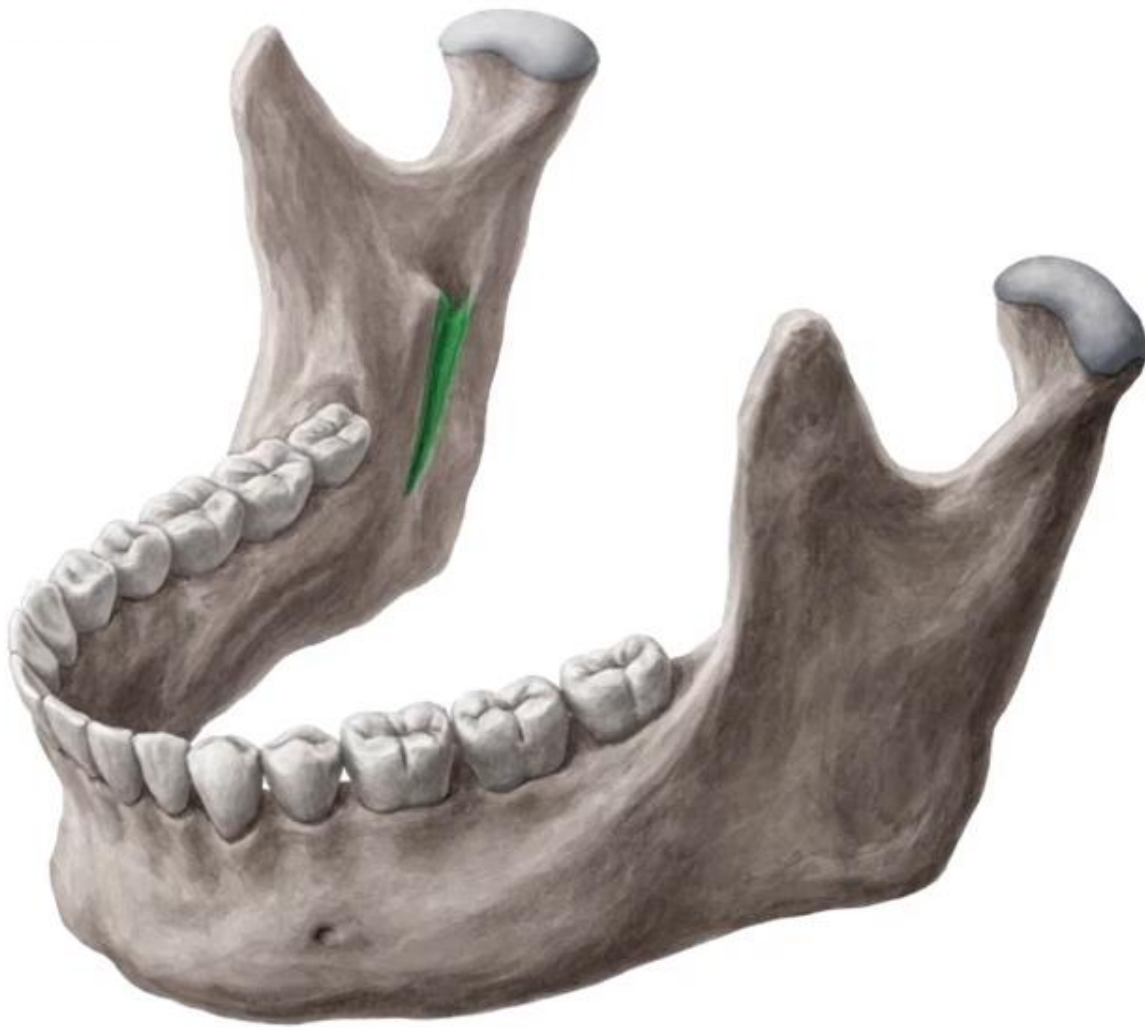
Mandibular foramen



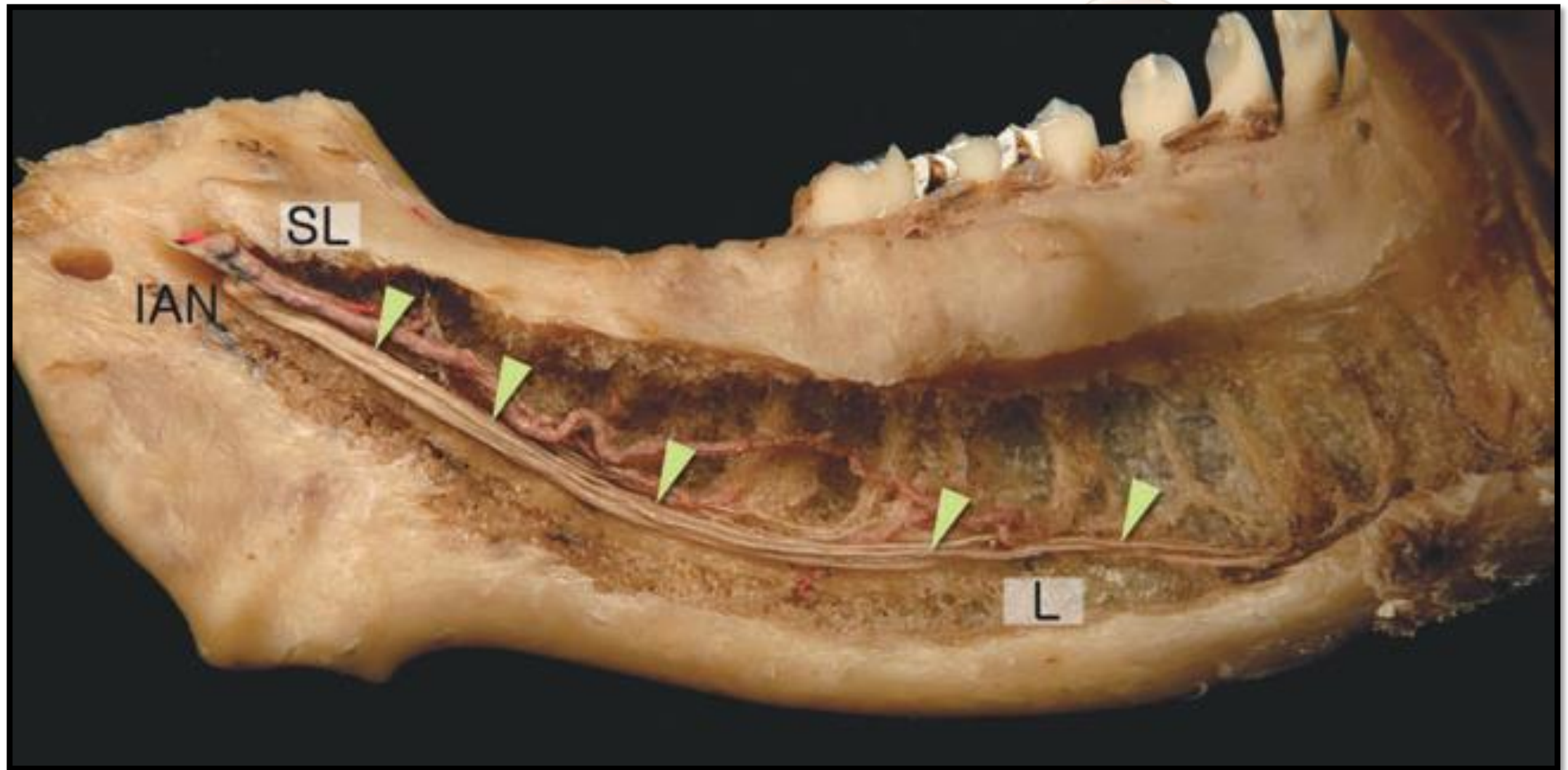
Lingula



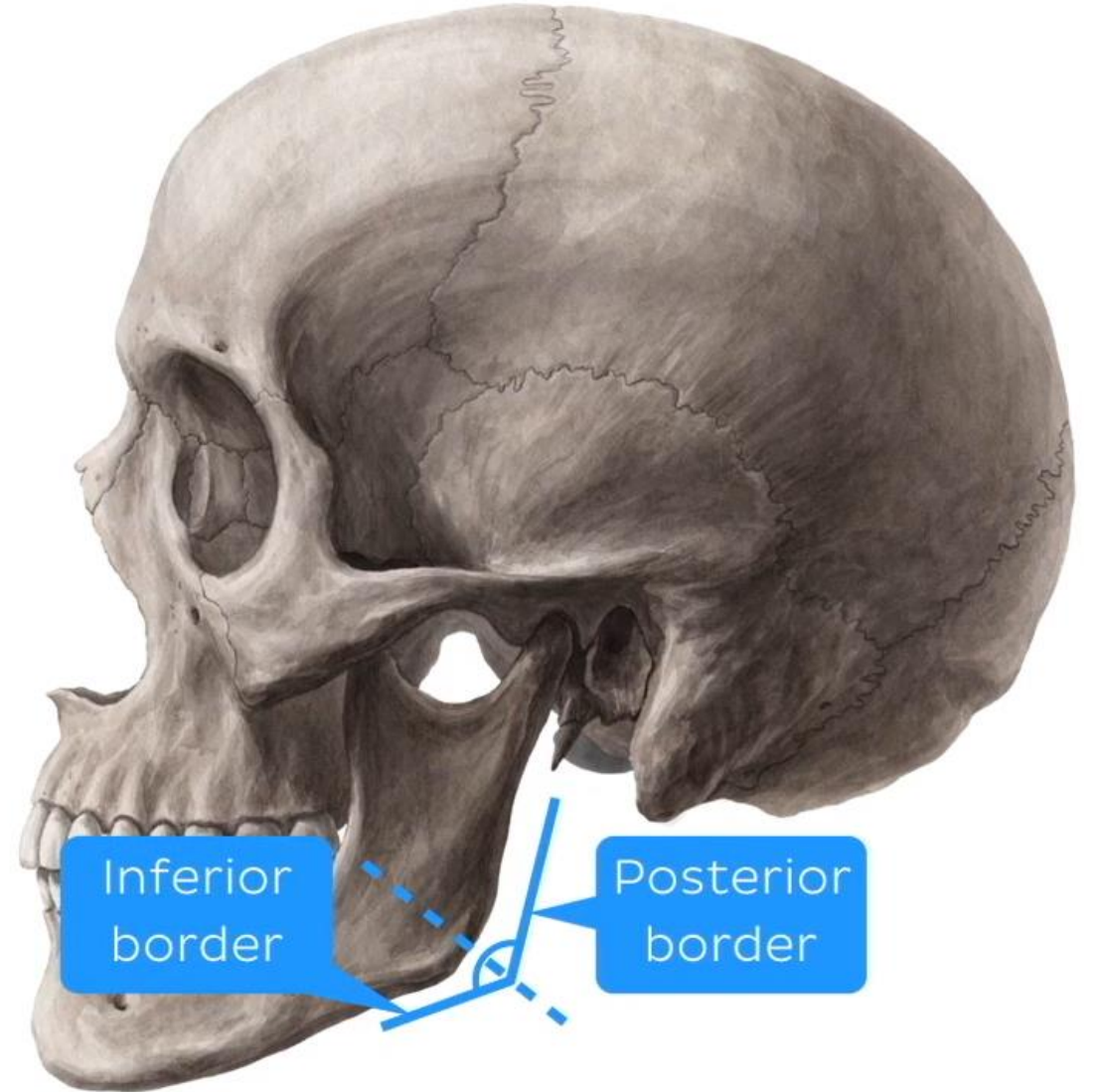
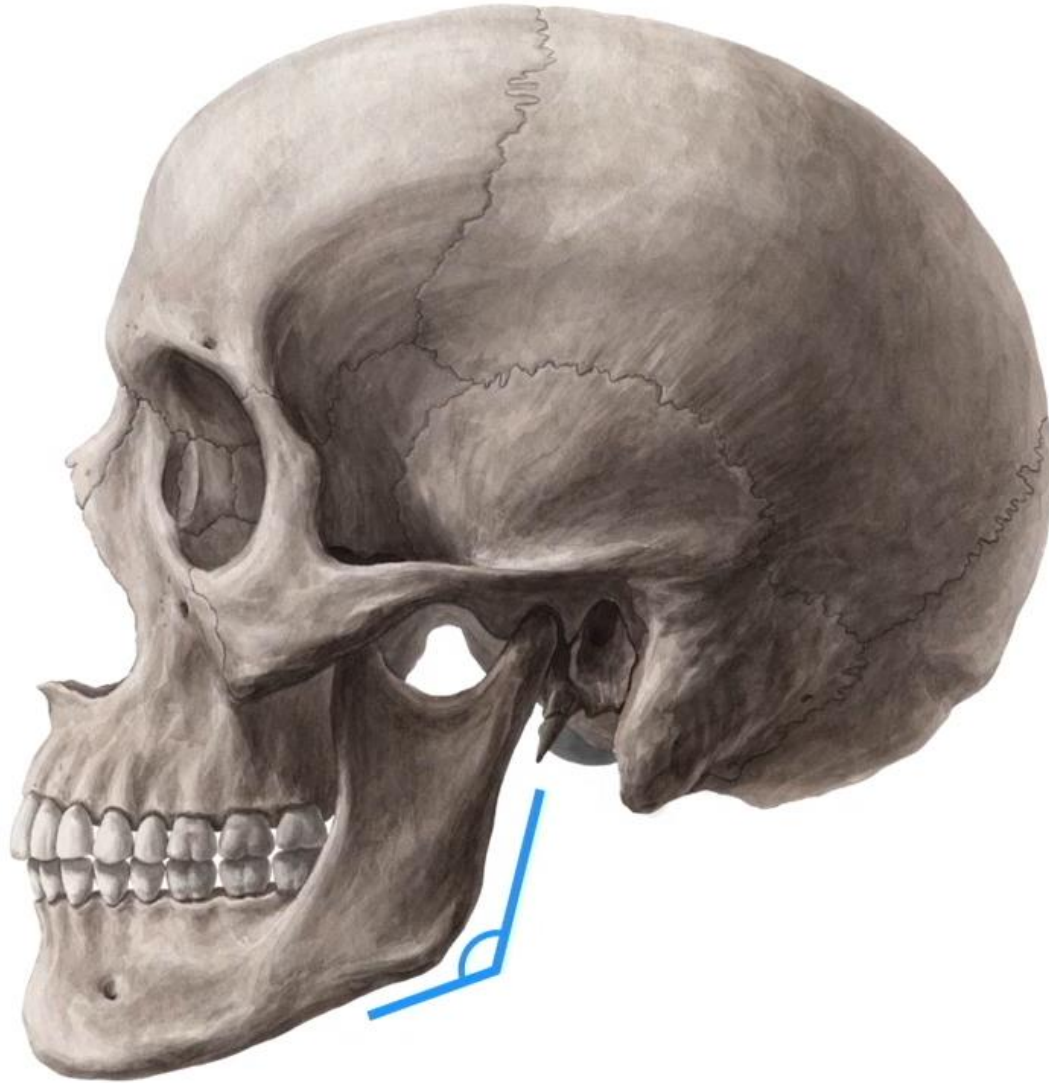
Mylohyoid groove

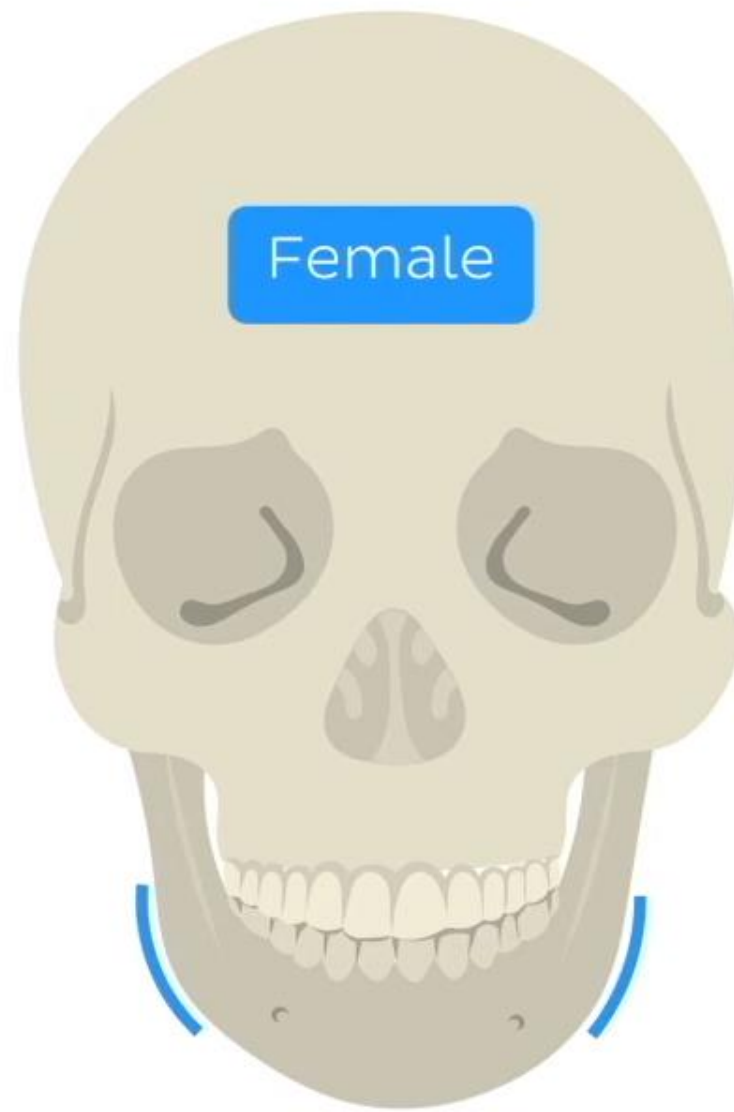
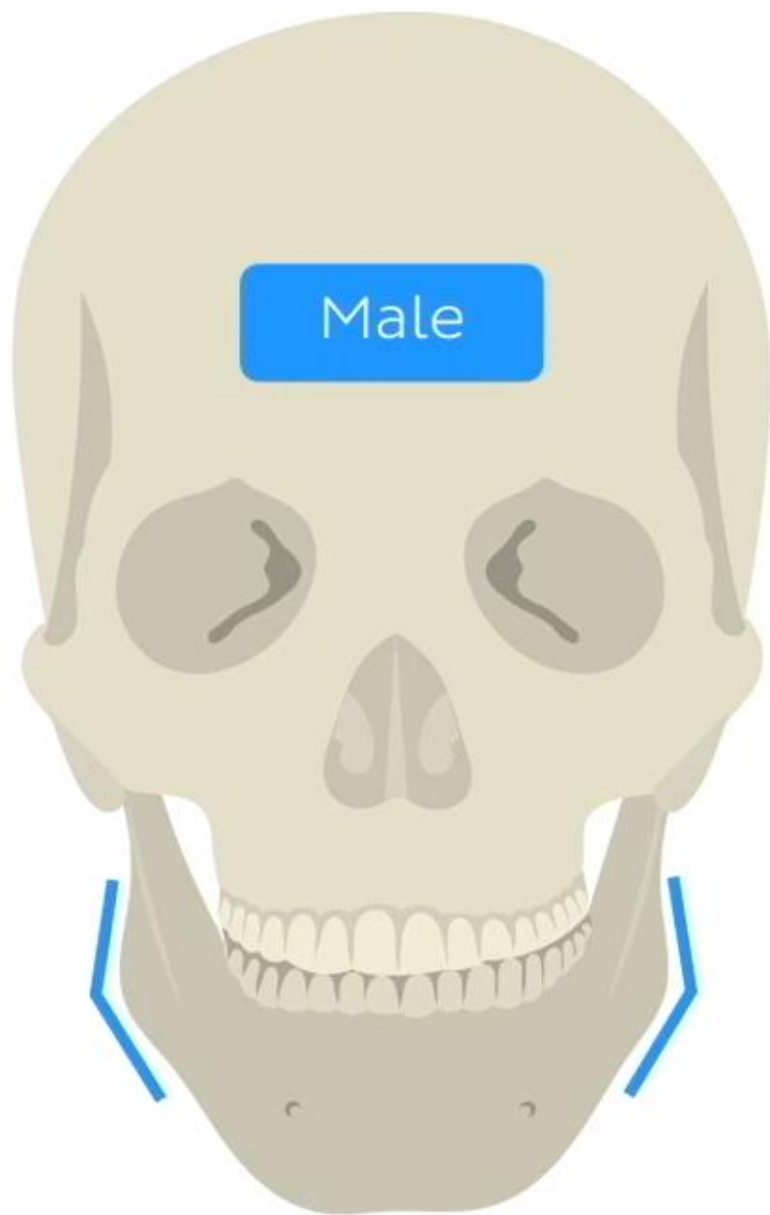


Mandibular canal

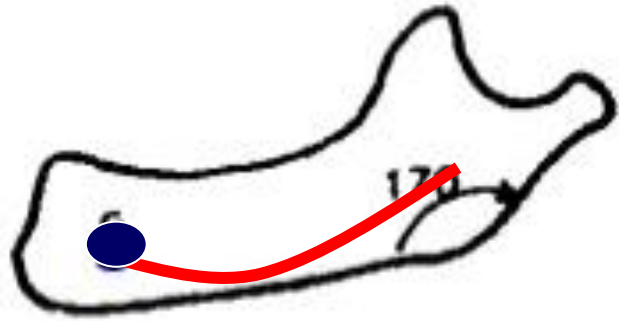


Angle of the mandible





Mandible at birth



The body of the mandible:

Small.

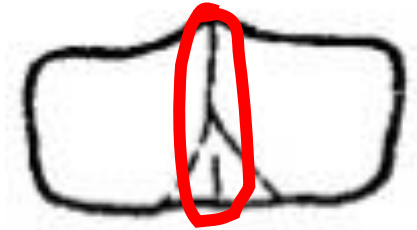
Consists of **two** separate **halves**.

Unite at the midline by fibrous tissue (the area of union called **symphysis menti**).

No teeth.

The mandibular canal runs near **the lower border** of the mandible.

The mental foramen is below **the crypt** of D.

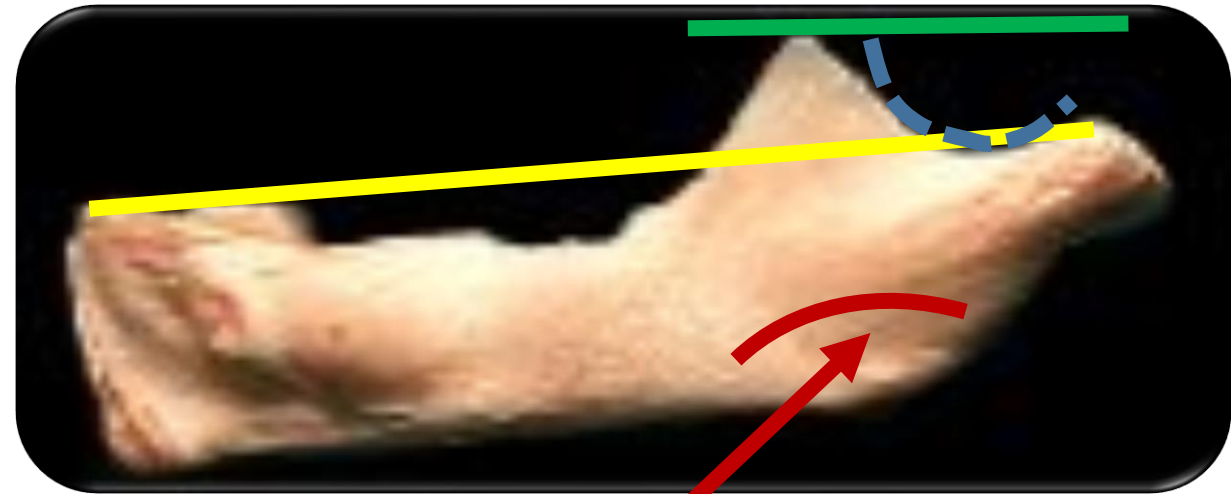


The ramus of the mandible at birth

The coronoid process

The sigmoid notch is **shallow**

The condyle



mandibular angle is **170**

Mandible at 3 years (after the eruption of all deciduous teeth)

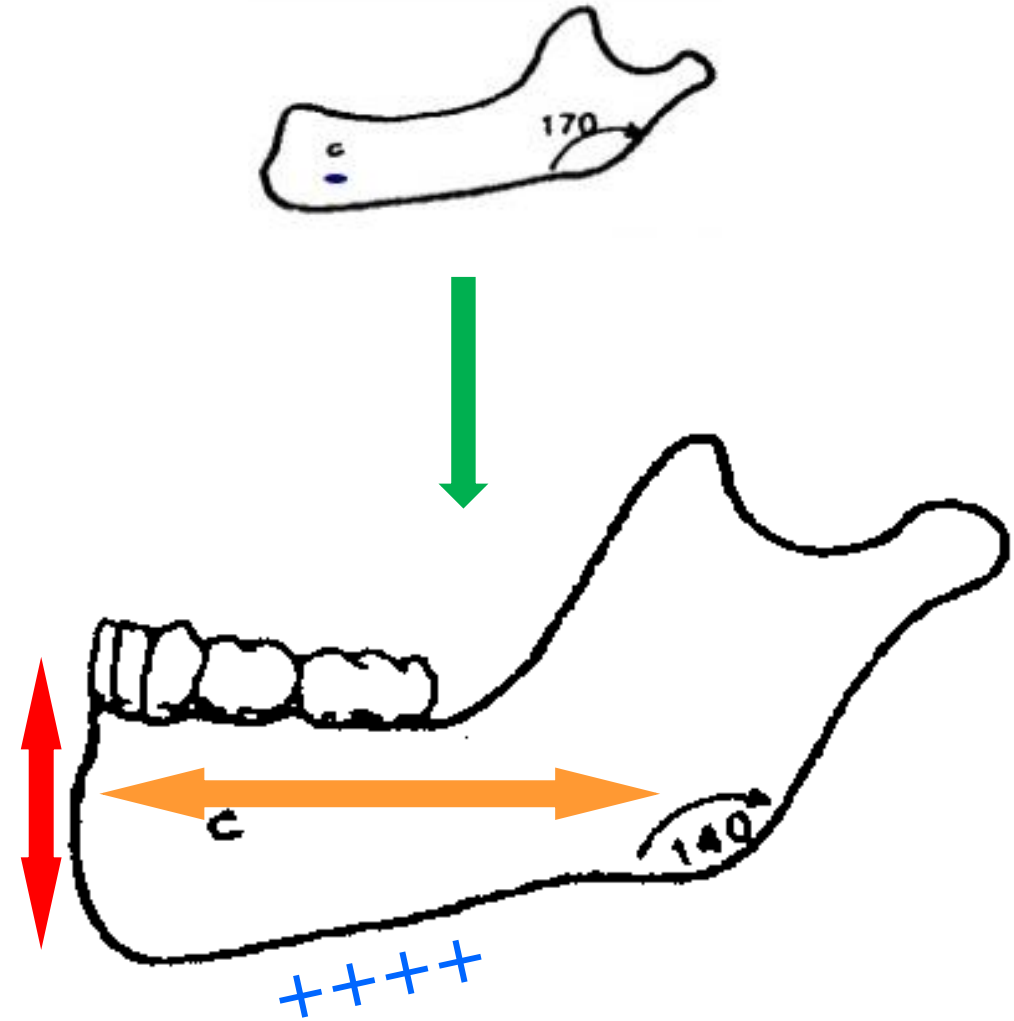


The two halves of mandible joint at the **symphysis menti** area from below upward by bone at the **end of 1st year.**

Mandible at 3 years (after the eruption of all deciduous teeth)

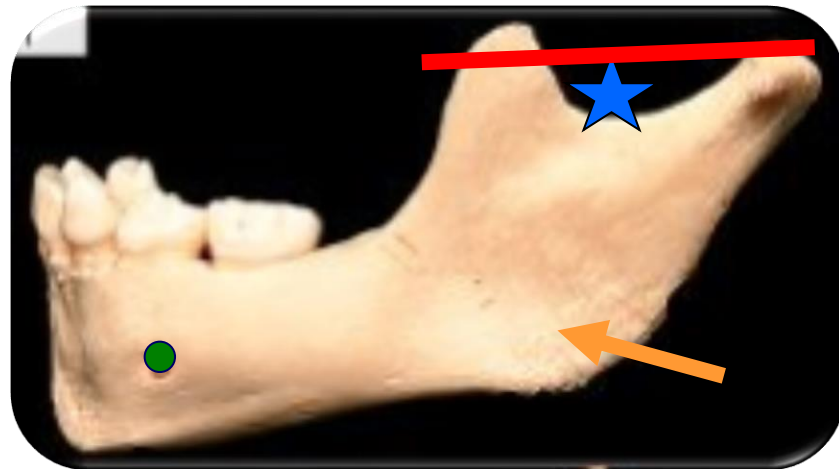
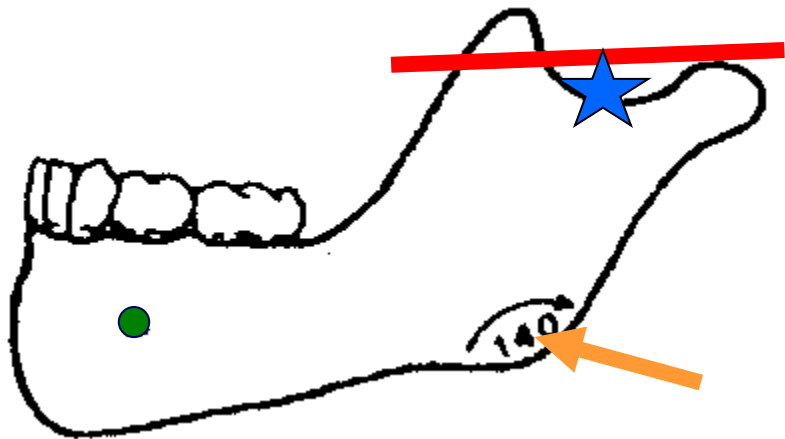
The body of the mandible:

- 1- **Elongates** especially behind mental foramen.
- 2- Increase **in height** by the growth of the alveolar process, accompany the eruption of all the deciduous teeth.
- 3- Bone deposition at the **lower border** of the mandible.



The ramus of the mandible at 3 years

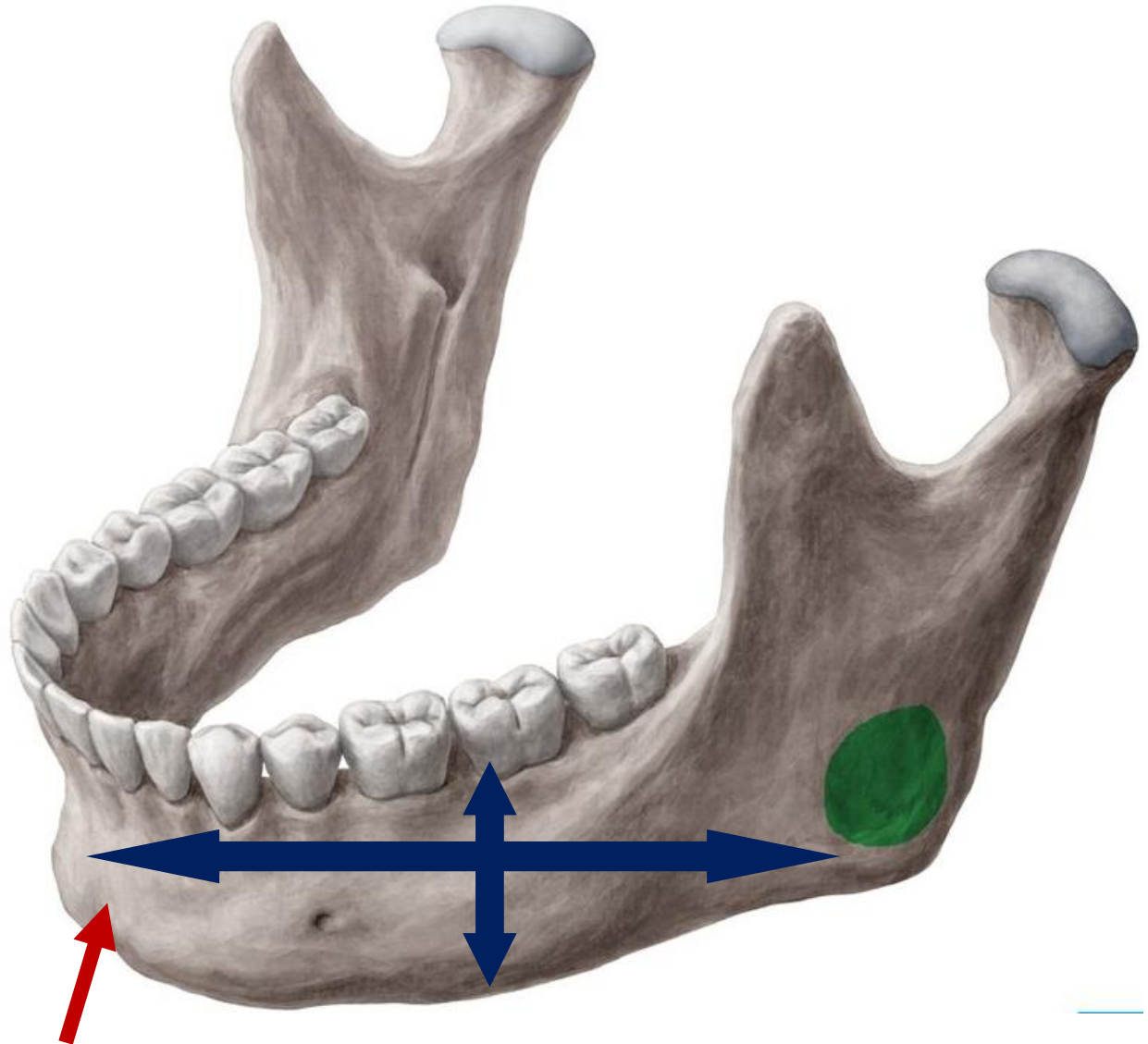
- 1-The **condyle** is at a **lower** level than the **coronoid** process
- 2-The **sigmoid notch** becomes deeper.
- 3-The **mental foramen** is below the socket of **lower D**.
- 4- Mandibular **angle** becomes **140**.



Mandible at adult age

Appearance of **mental protuberance**.

The mandible increases in **length and height**.

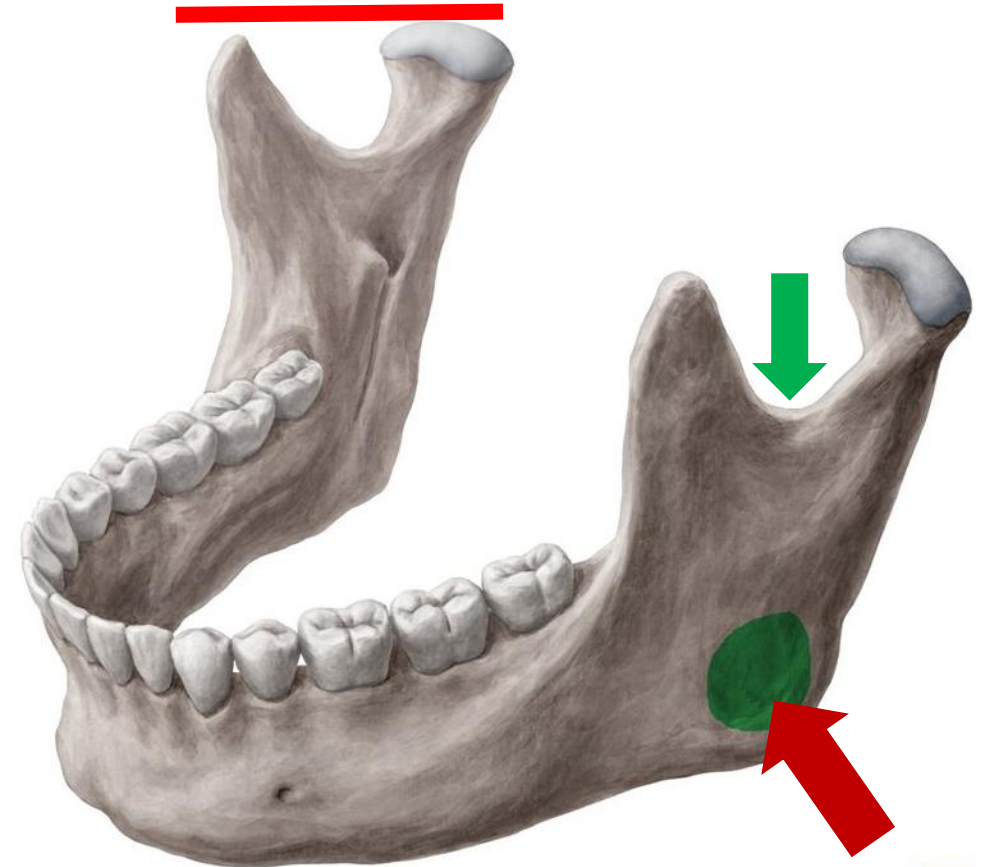
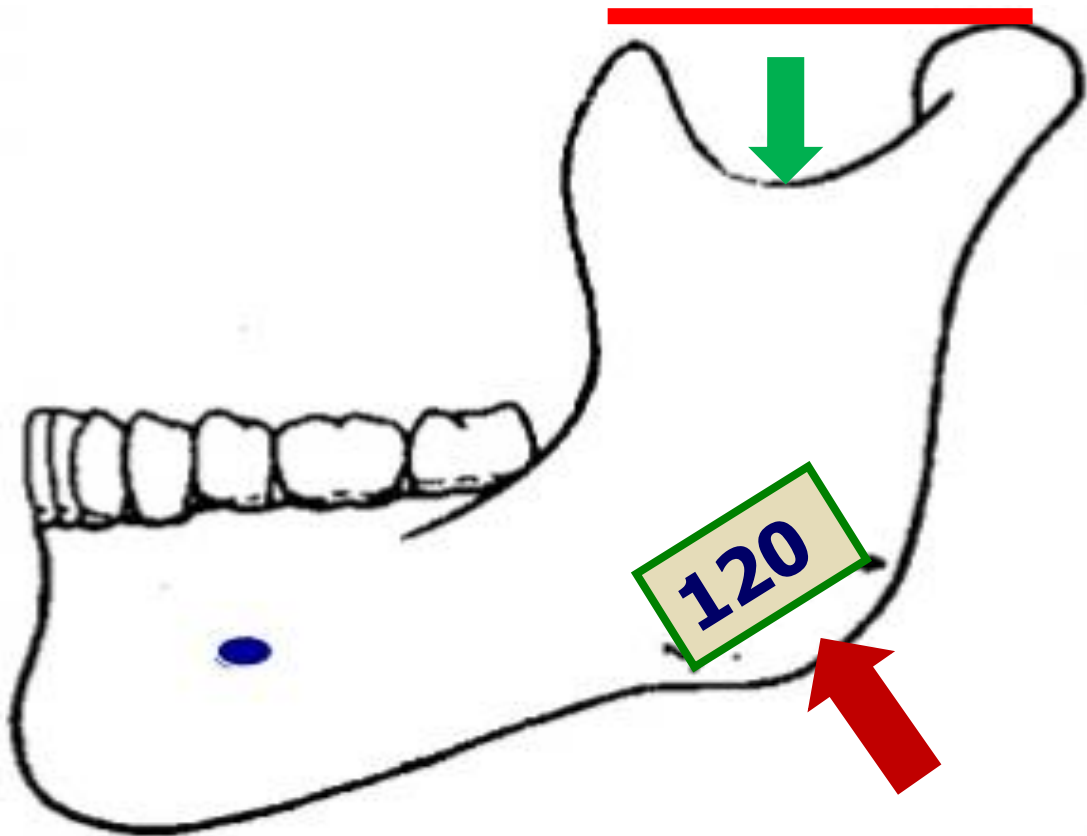


Mandible at adult age

3-The **condyle** present at a **higher** level than that of the **coronoid** process.

4-The **sigmoid notch** becomes **deeper**.

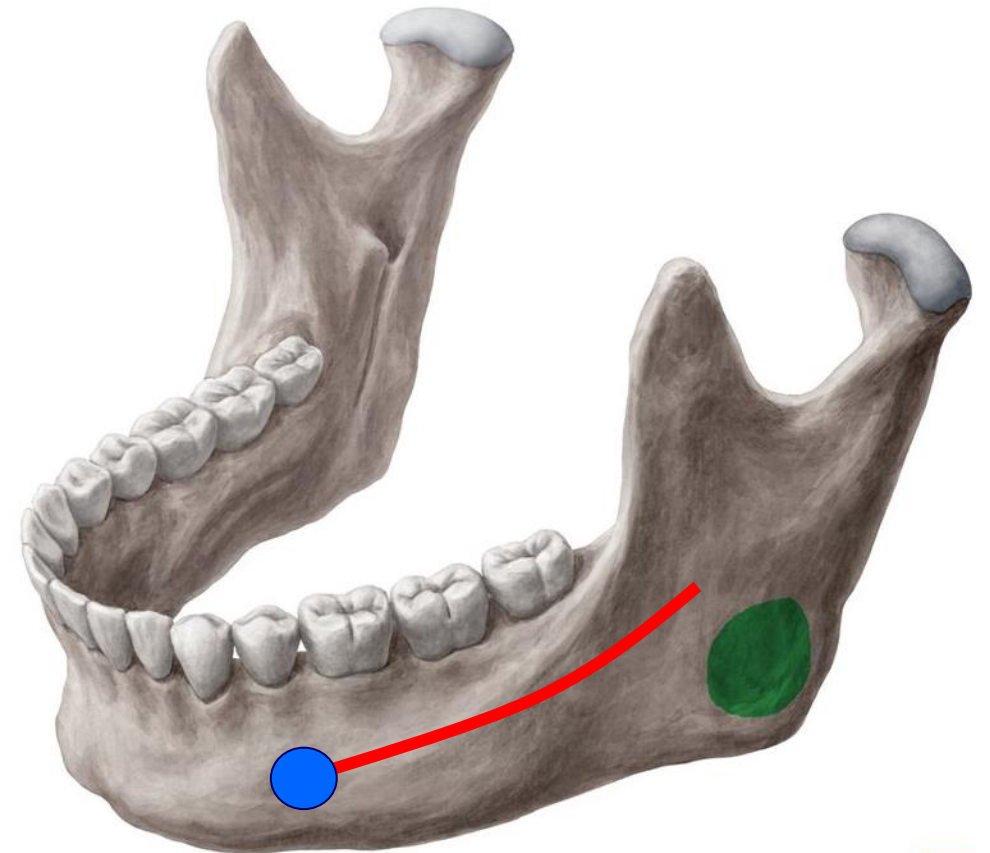
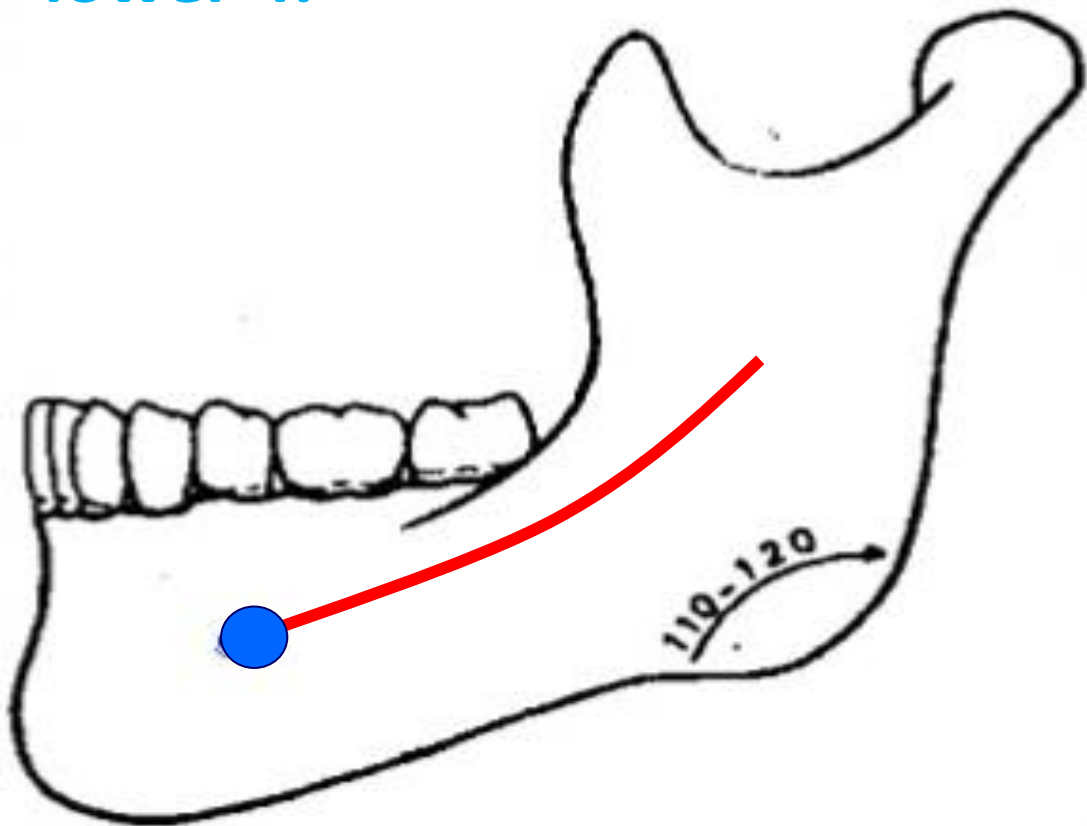
5-The mandibular **angle** becomes **110-120**.



Mandible at adult age

6-The **mandibular canal** runs parallel to the **mylohyoid Line**, midway between upper and lower borders.

7-The **mental foramen** is present below the socket of **lower 5 & lower 4**.



Mandible in old age



Natural teeth



After extraction



Resorption after 10 years



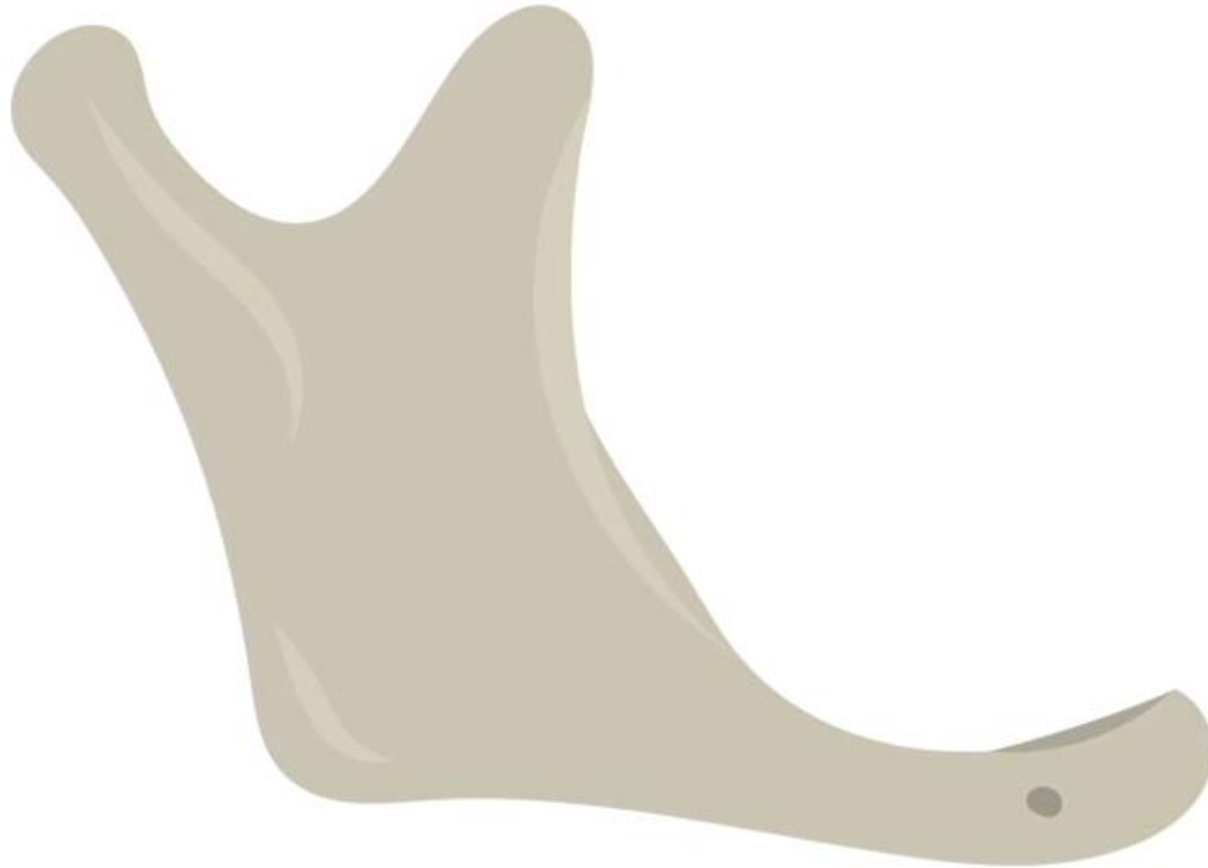
Resorption after 30 years

Following loss of the teeth



The mandible is **reduced in size** due to resorption of the alveolar bone.

Resorption of alveolar bone



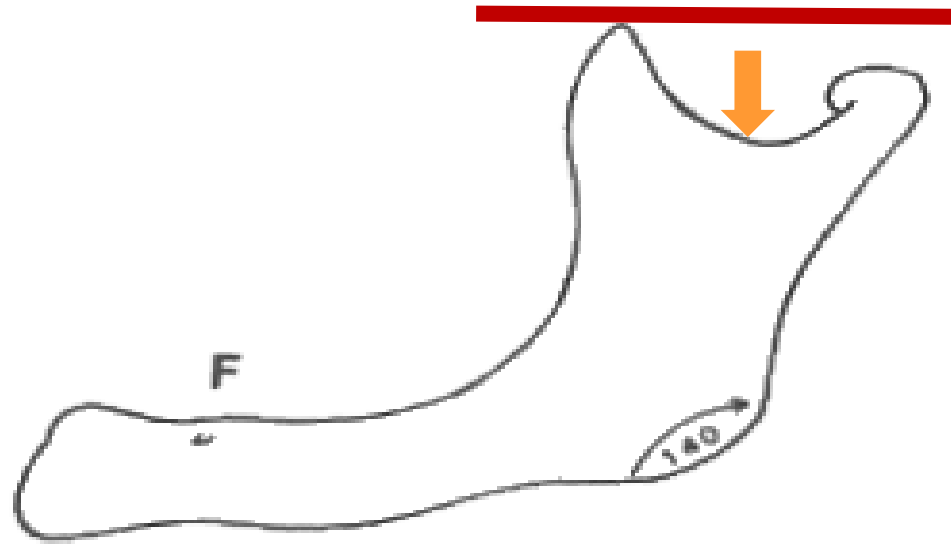
Mandible in old age



Mandible in old age

The **condyle** is more or less bent backward till it becomes in a **lower level** than the **coronoid process**.

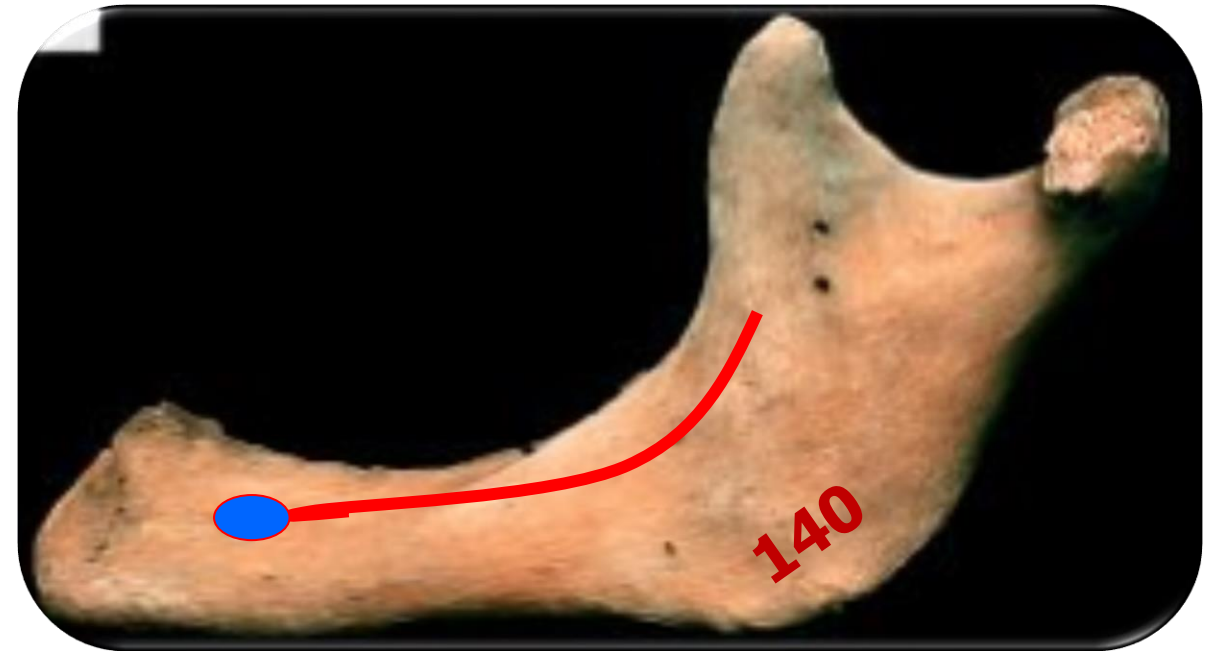
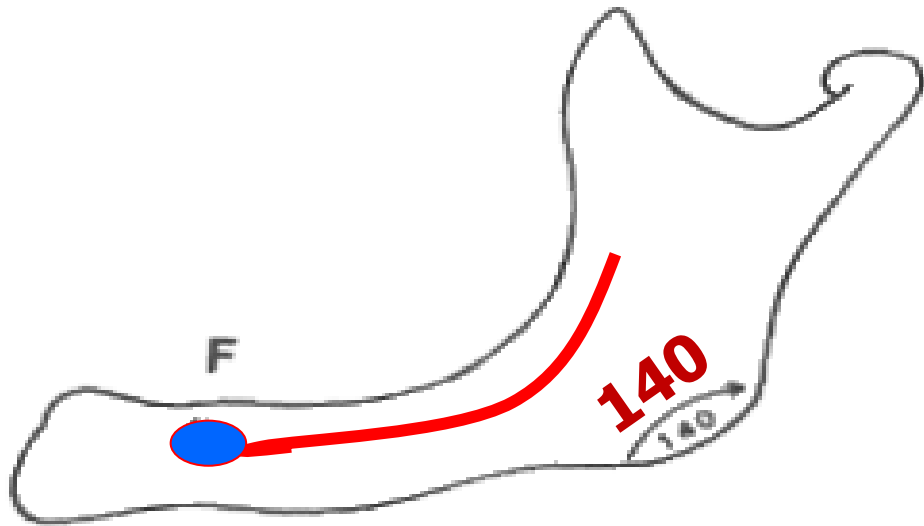
The **sigmoid notch** is shallower than in adult.



Mandible in old age

The **mandibular canal** and the **mental foramen** are close to the **upper border** of the body of mandible.

The ramus is oblique in direction with increasing in the mandibular angle **140**.



Thank You!

